From Beta to Q: CAPM as a Phase-Twisted Discounted Cash Flow Geometry

Reinterpreting CAPM risk-adjusted discounting as a complex-plane decomposition of admitted value, retained risk pressure, and market-implied residuals

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Abstract

The Capital Asset Pricing Model, or CAPM, is usually taught as a real-valued theory of expected return. A security's systematic market risk is compressed into beta, β, and beta is then converted into a required return through the equity risk premium:

$$r_{CAPM} = r_f + \beta(r_m - r_f). \quad (0.1)$$

When CAPM is used inside discounted cash-flow valuation, the future cash flow is discounted by this risk-adjusted rate:

$$R = \Sigma_t CF_t / (1 + r_{CAPM})^t. \quad (0.2)$$

This produces a single scalar present value. The scalar is useful, but it hides a structural question:

Where did the risk-adjusted part of the cash-flow amplitude go? (0.3)

This article proposes a geometric answer. Instead of treating CAPM only as a higher discount rate, we reinterpret the CAPM risk premium as a **phase angle**. A future cash flow is first discounted by a base rate to form a cash-flow amplitude A_t. CAPM's beta premium then rotates part of that amplitude away from the real axis. The ordinary CAPM DCF value becomes the real projection R_t, while the orthogonal complement becomes Q_t, a dollar-valued market-risk pressure coordinate.

The central identity is:

$$A_t^2 = R_t^2 + Q_t^2. \quad (0.4)$$

where:

$$A_t = CF_t / (1 + r_{base})^t. \quad (0.5)$$

$$R_t = A_t \cos \theta_t = CF_t / (1 + r_{CAPM})^t. \quad (0.6)$$

$$Q_t = A_t \sin \theta_t. \quad (0.7)$$

The CAPM risk angle is defined by:

$$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t. \quad (0.8)$$

Since:

$$r_{CAPM} = r_{base} + \beta ERP, \quad (0.9)$$

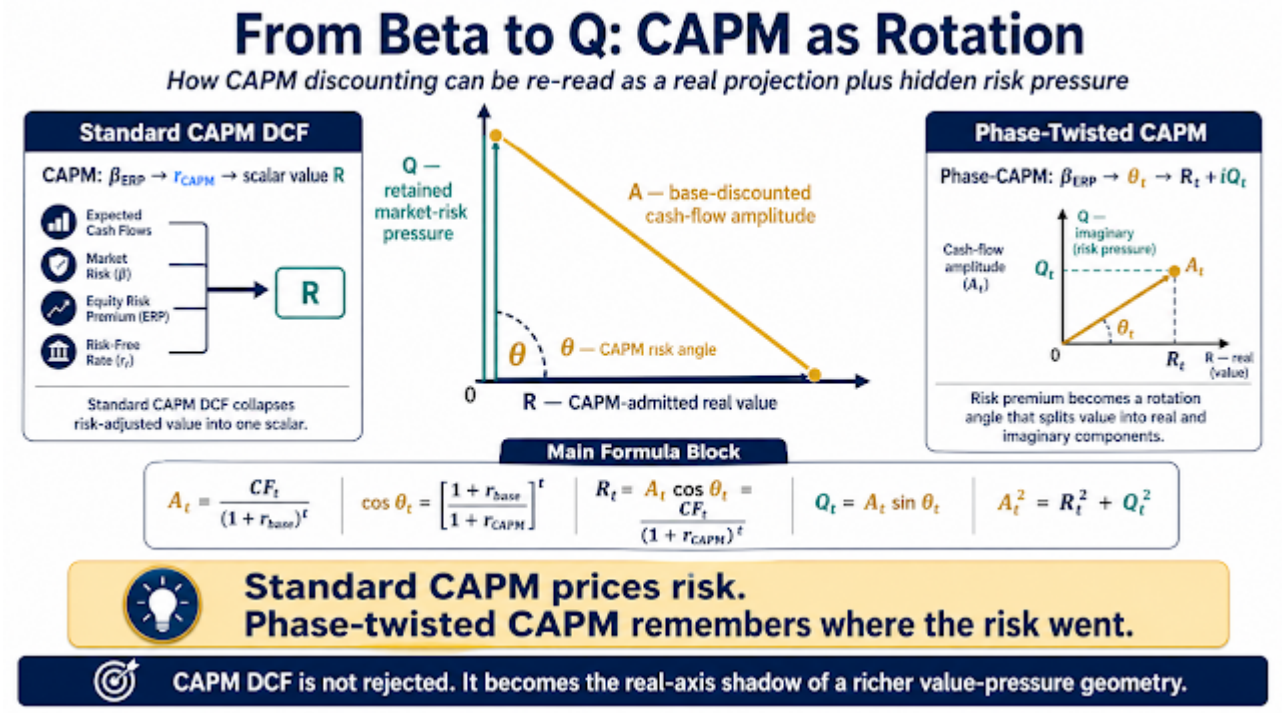
we also have:

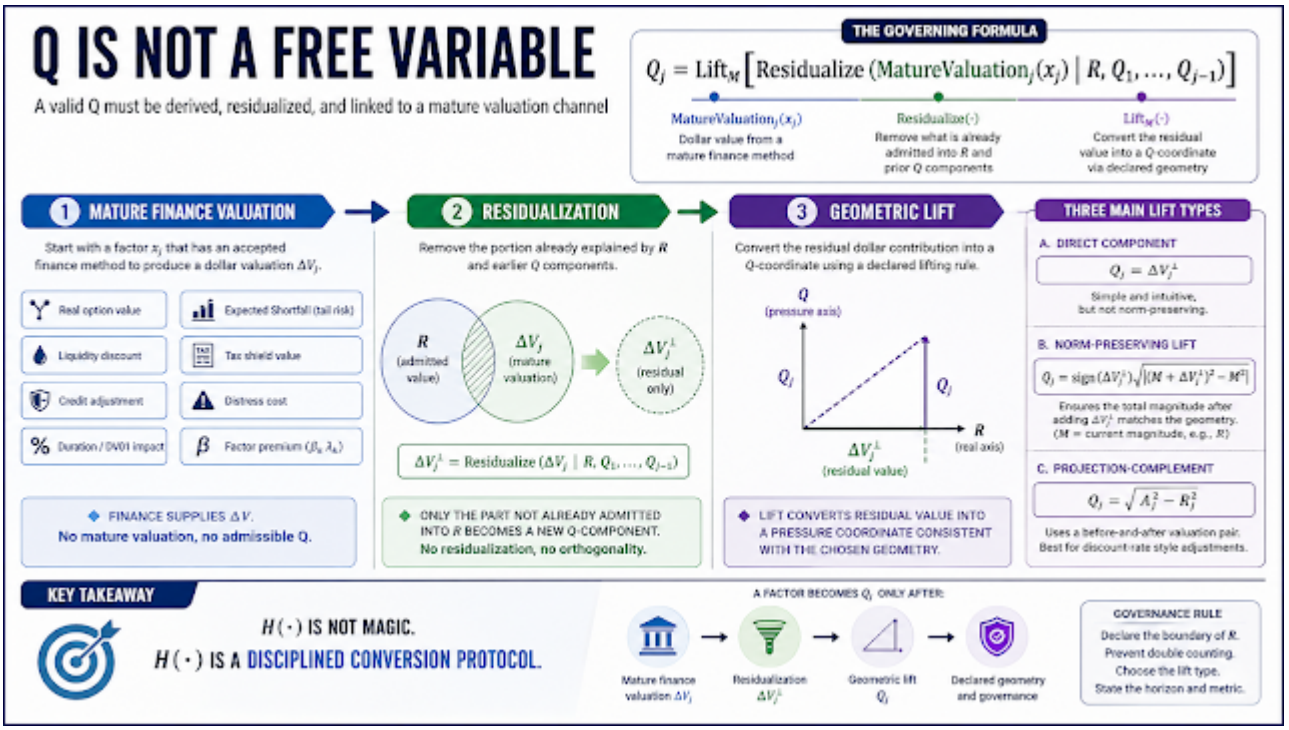
cos θ_t = [(1 + r_base)/(1 + r_base + βERP)]^t. (0.10)

This means Q is not an arbitrary new risk factor. It is a derived coordinate. It preserves information that ordinary scalar CAPM DCF throws away.

The purpose of this article is not to replace CAPM. It is to show that CAPM discounting can be re-read as a **projection geometry**:

CAPM prices risk. Phase-twisted CAPM remembers where the risk went. (0.11)

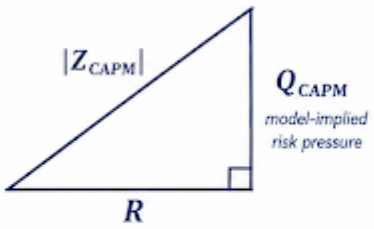




Reading Market Price Through R and Q

Separating admitted value from market-implied residual pressure.

Model Geometry

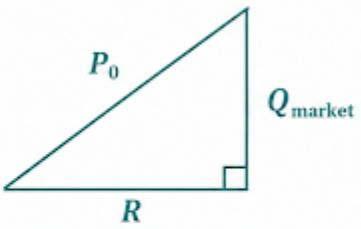


Q_{CAPM} = model-implied risk pressure

$$Q_{market} = \sqrt{P_0^2 - R^2}$$

$$\Delta Q = Q_{market} - Q_{CAPM}$$

Market Geometry



Condition	Interpretation
$\Delta Q \approx 0$	Market price fits CAPM phase geometry
$\Delta Q > 0$	Extra optionality, narrative premium, hidden demand, speculation
$\Delta Q < 0$	Market distrust, hidden distress, model too optimistic

Side Note

Price gap: $P_0 - R$

Pressure gap: $\Delta Q = Q_{market} - Q_{CAPM}$

Price above R is not automatically "overvaluation."
It may contain unresolved Q-like pressure.

The Q-framework turns market mispricing into a pressure diagnostic, not just a scalar price gap.

What Q Adds to Existing Finance Formulas

A comparison of scalar CAPM, DCF, certainty-equivalent valuation, and phase-twisted geometry

Framework	Main Formula	What Risk Does	What Q Adds
1 CAPM	$r_{CAPM} = r_f + \beta ERP$	Converts beta into required return	Q not needed for SML itself
2 Risk-Adjusted DCF	$R = \sum_{t=1}^{\infty} \frac{CF_t}{(1+r_{CAPM})^t}$	Reduces scalar present value	Preserves hidden complement
3 Certainty Equivalent	$R_t = \frac{CE_t}{(1+r_{base})^t}$	Converts risky CF into safer CF	Reveals CE as real projection
4 Phase-Twisted DCF	$A_t^2 = R_t^2 + Q_t^2$	Rotates value amplitude	Shows risk pressure explicitly
5 Market-Calibrated Q	$P_0^2 = R^2 + Q_{market}^2$	Market implies residual pressure	Enables ΔQ diagnostic

Scalar value (Single-number valuation) → **Projection geometry** (Amplitude + complement) → **Pressure diagnostics** (Market-implied residuals)

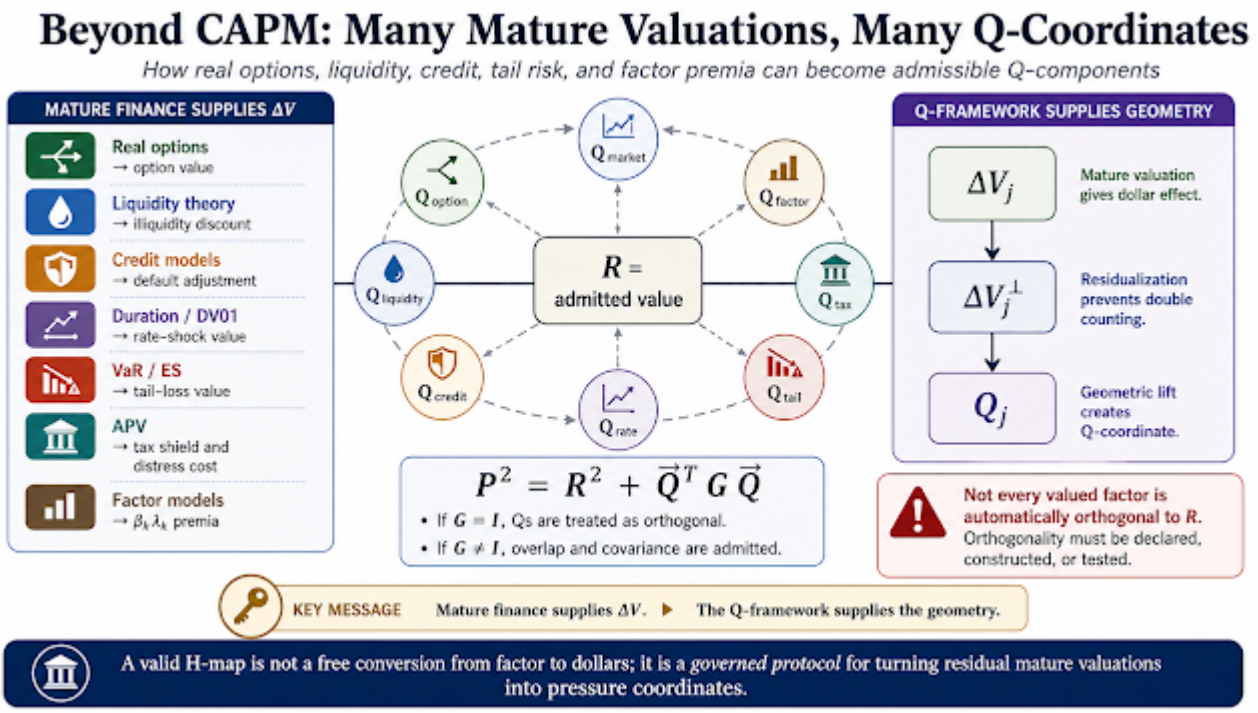
$$CE_t = CF_t \cos\theta_t \qquad Q_t = \left[\frac{CF_t}{(1+r_{base})^t}\right] \sin\theta_t$$

Key Message

Q does not replace mature finance.

Q organizes what mature finance already values.

The advantage of Q is not more complexity for its own sake, but **cleaner decomposition** of *value*, pressure, and residuals.



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0. Reader's Guide: What This Article Is and Is Not

0.1 What this article is

This article is a teaching and conceptual article. It develops a complex-plane extension of CAPM-based discounted cash-flow valuation.

It starts from an earlier idea: a complex economic state can be written as:

Z = R + iP. (0.12)

where R represents visible or ledgered value, while P represents unresolved pressure or protection that is not yet ledgered as consequence. The earlier tutorial also treats rotation as a change of frame: the same economic state is re-read through valuation, stress testing, settlement, admissibility, risk control, or protection.

This article takes that general grammar and applies it more tightly to CAPM.

The goal is to answer a precise question:

Can CAPM risk-adjusted discounting be re-expressed as a clean real/imaginary decomposition? (0.13)

The proposed answer is yes.

0.2 What this article is not

This article is **not** investment advice.

It is **not** a claim that CAPM is wrong.

It is **not** a claim that Q is already a standard CAPM variable.

It is **not** a claim that markets literally obey complex-number physics.

It is **not** a replacement for:

CAPM,

DCF,

certainty-equivalent valuation,

Black–Scholes,

mean–variance analysis,

factor models,

portfolio optimization,

VaR,

Expected Shortfall,

or accounting valuation rules.

The safer claim is:

Phase-twisted CAPM is a coordinate transformation and diagnostic extension of CAPM DCF. (0.14)

It does not change the ordinary CAPM DCF value. It explains that value as a real-axis projection and preserves the orthogonal complement as Q.

0.3 The discipline rule

Adding new variables is dangerous.

A weak framework invents symbols because they look elegant.

A stronger framework adds a variable only when the variable preserves structure, simplifies a relationship, or exposes a diagnostic that the old notation hides.

Therefore this article follows one discipline rule:

A new variable is justified only if it preserves information that the old scalar formula hides. (0.15)

The new variable Q must therefore earn its place.

It will do so through relationships such as:

$A_t^2 = R_t^2 + Q_t^2$. (0.16)

$Q_t/R_t = \tan \theta_t$. (0.17)

$Loss_t \approx Q_t^2/(2A_t)$. (0.18)

$\Delta Q = Q_{market} - Q_{CAPM}$. (0.19)

These are not decorative formulas. They show that Q can turn CAPM discounting into a cleaner geometry of value, risk pressure, market-implied residuals, and duration sensitivity.

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1. The Problem: CAPM Is Elegant but Scalar

1.1 The standard CAPM relation

The standard CAPM relation is:

$r_{CAPM} = r_f + \beta(r_m - r_f)$. (1.1)

where:

r_f = risk-free rate. (1.2)

r_m = expected market return. (1.3)

$r_m - r_f$ = equity risk premium. (1.4)

β = systematic market-risk loading. (1.5)

It is convenient to write:

$ERP = r_m - r_f$. (1.6)

Then:

$r_{CAPM} = r_f + \beta ERP$. (1.7)

For this article, we use a slightly generalized base-rate notation:

$r_{CAPM} = r_{base} + \beta ERP$. (1.8)

Usually r_{base} may be r_f. But it can also be another declared baseline rate, provided the valuation protocol is explicit.

1.2 CAPM inside DCF

Once the CAPM discount rate is chosen, the ordinary risk-adjusted DCF value is:

$R = \Sigma_t CF_t / (1 + r_{CAPM})^t$. (1.9)

For one cash-flow component:

$R_t = CF_t / (1 + r_{CAPM})^t$. (1.10)

This is mathematically clean.

But it compresses the whole operation into one real number.

The cash flow begins as a future value:

CF_t. (1.11)

Then market risk enters through beta:

βERP. (1.12)

Then the result becomes one scalar present value:

R_t. (1.13)

The path is:

CF_t + βERP + t → R_t. (1.14)

Something has been simplified. That is useful. But something has also been hidden.

1.3 What gets hidden

Suppose we compare two valuations.

Asset A has high base cash-flow strength but also high CAPM risk.

Asset B has lower base cash-flow strength but lower CAPM risk.

Both may produce the same CAPM present value R.

In ordinary scalar DCF, they may collapse to the same number:

R_A = R_B. (1.15)

But their internal geometry is different.

Asset A may have:

large amplitude, large risk-pressure rotation. (1.16)

Asset B may have:

smaller amplitude, smaller risk-pressure rotation. (1.17)

A scalar present value does not preserve that difference.

That is the central motivation for Q.

The question is not:

Can we make CAPM more complicated? (1.18)

The question is:

Can we preserve the value-pressure structure that CAPM DCF compresses? (1.19)

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2. Why Inventing Q Is Dangerous Unless It Clarifies CAPM

2.1 The bad way to add Q

The bad version would be:

R = CAPM DCF value. (2.1)

Q = some extra volatility number. (2.2)

That is dangerous because CAPM has already used beta to price systematic market risk.

If Q is then created from another version of the same market risk, the model may double count risk.

The result would be:

risk appears once in R through r_CAPM; risk appears again in Q through σ. (2.3)

That is not a clean orthogonal decomposition.

It is just overlapping notation.

2.2 The good way to add Q

The good version is different.

Risk does not enter twice.

Instead:

βERP defines θ_t. (2.4)

θ_t splits A_t into R_t and Q_t. (2.5)

So the logical chain is:

βERP → θ_t → R_t + iQ_t. (2.6)

This means Q is not an extra independent assumption.

Q is the orthogonal complement produced by the same CAPM risk premium.

In other words:

CAPM does not give R and then Q is added. (2.7)

Rather:

CAPM gives a risk angle, and the risk angle decomposes amplitude into R and Q. (2.8)

That is the key move.

2.3 The required test

If Q is worth adding, it should simplify at least one existing relationship.

The proposed framework gives several candidates.

First:

$A_t^2 = R_t^2 + Q_t^2$. (2.9)

This says CAPM discounting can be read as Pythagorean decomposition.

Second:

$Q_t/R_t = \tan \theta_t$. (2.10)

This says beta risk can be read as a real-to-imaginary value tilt.

Third:

$Loss_t \approx Q_t^2/(2A_t)$. (2.11)

This says the CAPM value haircut is approximately quadratic in risk pressure.

Fourth:

$\Delta Q = Q_{market} - Q_{CAPM}$. (2.12)

This says the gap between market price and CAPM geometry can be measured as residual pressure rather than only price error.

These relationships are the reason Q may be worth adding.

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3. Base-Discounted Cash Flow as Amplitude

3.1 Define the base-discounted amplitude

Let CF_t be the expected cash flow at time t.

Instead of immediately discounting it by r_{CAPM}, first discount it by a base rate:

$A_t = CF_t / (1 + r_{base})^t$. (3.1)

This A_t is the **base-discounted cash-flow amplitude**.

It is the present value of the cash flow before CAPM market-risk projection.

If $r_{base} = r_f$, then A_t is the risk-free discounted cash-flow amount.

If r_{base} is another declared baseline, then A_t is the cash-flow amplitude under that declared baseline.

The article does not require r_{base} to always equal r_f. But it does require that r_{base} be declared.

No declared baseline, no meaningful A_t. (3.2)

3.2 Why call A_t an amplitude?

In ordinary DCF, CF_t becomes a present value after discounting.

In the phase-twisted model, we separate two operations:

base discounting;

risk rotation.

So the path becomes:

$CF_t \rightarrow A_t \rightarrow R_t + iQ_t$. (3.3)

A_t is called an amplitude because it is not yet split into real admitted value and imaginary risk pressure.

It is the pre-CAPM-projection value magnitude.

The ordinary CAPM DCF value R_t will become the real projection of A_t.

The new Q_t will become the imaginary complement of that projection.

This is why A_t is not merely another present value. It is the object from which both R_t and Q_t are derived.

3.3 The first geometric picture

Imagine A_t as the length of a vector.

The horizontal axis is the real value axis.

The vertical axis is the risk-pressure axis.

The vector A_t is tilted by an angle θ_t.

Then:

$R_t = A_t \cos \theta_t$. (3.4)

$Q_t = A_t \sin \theta_t$. (3.5)

Therefore:

$A_t^2 = R_t^2 + Q_t^2$. (3.6)

This is only geometry so far. The important question is:

Can θ_t be defined from CAPM itself? (3.7)

The answer is yes.

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4. CAPM Risk Premium as a Phase Angle

4.1 Define the CAPM risk angle

We want the real projection R_t to equal the ordinary CAPM DCF value.

So we require:

$R_t = CF_t / (1 + r_{CAPM})^t$. (4.1)

But we also want:

$$R_t = A_t \cos \theta_t. \quad (4.2)$$

Since:

$$A_t = CF_t / (1 + r_{base})^t, \quad (4.3)$$

we need:

$$[CF_t / (1 + r_{base})^t] \cos \theta_t = CF_t / (1 + r_{CAPM})^t. \quad (4.4)$$

Cancel CF_t:

$$\cos \theta_t / (1 + r_{base})^t = 1 / (1 + r_{CAPM})^t. \quad (4.5)$$

Therefore:

$$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t. \quad (4.6)$$

Since:

$$r_{CAPM} = r_{base} + \beta ERP, \quad (4.7)$$

we get:

$$\cos \theta_t = [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^t. \quad (4.8)$$

This is the central definition of the CAPM risk angle.

4.2 Interpretation

In ordinary CAPM DCF, the beta premium raises the discount rate.

In phase-twisted CAPM, the beta premium creates a rotation angle.

The two are mathematically linked.

The old sentence is:

Higher βERP reduces present value. (4.9)

The new sentence is:

Higher βERP rotates more of the base-discounted amplitude away from the real axis. (4.10)

This is not a different numerical valuation for R_t.

It is a different interpretation of how the valuation is formed.

The ordinary CAPM value is still preserved:

$$R_t = CF_t / (1 + r_{CAPM})^t. \quad (4.11)$$

But now we also preserve:

$$Q_t = A_t \sin \theta_t. \quad (4.12)$$

The scalar formula gives the real shadow.

The complex formula gives the projected state.

4.3 Small-risk approximation

When βERP is small relative to 1 + r_base, the angle θ_t is small.

Let:

$x = \beta ERP / (1 + r_{base})$. (4.13)

Then:

$\cos \theta_t = [1/(1 + x)]^t$. (4.14)

For small x:

$[1/(1 + x)]^t \approx 1 - tx$. (4.15)

For small θ_t:

$\cos \theta_t \approx 1 - \theta_t^2/2$. (4.16)

So:

$1 - \theta_t^2/2 \approx 1 - t\beta ERP/(1 + r_{base})$. (4.17)

Therefore:

$\theta_t^2/2 \approx t\beta ERP/(1 + r_{base})$. (4.18)

Thus:

$\theta_t \approx \sqrt{[2t\beta ERP/(1 + r_{base})]}$. (4.19)

Equivalently:

$\beta ERP \approx (1 + r_{base})\theta_t^2/(2t)$. (4.20)

This is a useful result.

CAPM is linear in beta premium:

βERP. (4.21)

But phase-CAPM is approximately quadratic in angle:

$\beta ERP \propto \theta_t^2$. (4.22)

That gives us a new geometric reading:

Market risk premium behaves like squared phase displacement. (4.23)

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5. Ordinary CAPM DCF as the Real Projection

5.1 Define real projection

Now define:

$R_t = A_t \cos \theta_t$. (5.1)

Substitute the two definitions:

$A_t = CF_t / (1 + r_{base})^t$. (5.2)

$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t$. (5.3)

Then:

$R_t = [CF_t / (1 + r_{base})^t] \cdot [(1 + r_{base})/(1 + r_{CAPM})]^t$. (5.4)

The factors $(1 + r_{base})^t$ cancel:

$R_t = CF_t / (1 + r_{CAPM})^t$. (5.5)

This proves the key point.

Ordinary CAPM DCF is exactly the real projection of the phase-twisted cash-flow amplitude. (5.6)

5.2 Why this matters

This result matters because it protects the framework from becoming arbitrary.

We are not saying:

Invent Q, then modify CAPM. (5.7)

We are saying:

Start from CAPM DCF, then reveal the projection geometry already implicit in it. (5.8)

The real part is not changed.

The ordinary CAPM valuation remains intact.

The difference is that the new framework refuses to erase the orthogonal complement.

That complement is Q.

5.3 The first main theorem

We can state the result as a theorem.

Theorem 1 — CAPM DCF projection theorem

Given:

$A_t = CF_t / (1 + r_{base})^t$. (5.9)

$r_{CAPM} = r_{base} + \beta ERP$. (5.10)

$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t$. (5.11)

Then:

$A_t \cos \theta_t = CF_t / (1 + r_{CAPM})^t$. (5.12)

Therefore, ordinary CAPM DCF is the real-axis projection of a base-discounted cash-flow amplitude rotated by the CAPM risk angle.

In compact form:

CAPM DCF = Re[phase-twisted DCF]. (5.13)

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6. Defining Q: The CAPM-Implied Risk Pressure Coordinate

6.1 Definition of Q_t

Once R_t is defined as the real projection, define Q_t as the imaginary projection:

$Q_t = A_t \sin \theta_t$. (6.1)

Then:

$Z_t = R_t + iQ_t$. (6.2)

and:

$Z_t = A_t(\cos \theta_t + i \sin \theta_t)$. (6.3)

Using Euler form:

$Z_t = A_t e^{(i\theta_t)}$. (6.4)

This is the one-period complex cash-flow state.

6.2 Pythagorean identity

Because:

$R_t = A_t \cos \theta_t$. (6.5)

$Q_t = A_t \sin \theta_t$. (6.6)

we have:

$R_t^2 + Q_t^2 = A_t^2\cos^2\theta_t + A_t^2\sin^2\theta_t$. (6.7)

So:

$R_t^2 + Q_t^2 = A_t^2(\cos^2\theta_t + \sin^2\theta_t)$. (6.8)

Since:

$\cos^2\theta_t + \sin^2\theta_t = 1$, (6.9)

we get:

$A_t^2 = R_t^2 + Q_t^2$. (6.10)

This is the core identity.

It means:

base-discounted cash-flow amplitude² = CAPM-admitted value² + CAPM-implied risk pressure². (6.11)

6.3 Explicit CAPM formula for Q_t

From:

$Q_t = A_t \sin \theta_t$, (6.12)

and:

$\sin \theta_t = \sqrt{(1 - \cos^2\theta_t)}$, (6.13)

we get:

$Q_t = A_t \sqrt{(1 - \cos^2\theta_t)}$. (6.14)

Substitute:

$A_t = CF_t / (1 + r_{base})^t$, (6.15)

and:

$\cos \theta_t = [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^t$. (6.16)

Then:

$Q_t = [CF_t/(1 + r_{base})^t] \cdot \sqrt{\{1 - [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^{(2t)}\}}$. (6.17)

This is the CAPM-derived H-function.

So we may write:

$Q_t = H_t(CF_t, r_{base}, \beta, ERP, t)$. (6.18)

where:

$H_t(CF_t, r_{base}, \beta, ERP, t) = [CF_t/(1 + r_{base})^t] \cdot \sqrt{\{1 - [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^{(2t)}\}}$. (6.19)

This is important.

Q is not a free variable.

Q is a derived CAPM risk-pressure coordinate.

6.4 What Q means

Q_t is not ordinary volatility.

Q_t is not beta itself.

Q_t is not alpha.

Q_t is not VaR.

Q_t is not Expected Shortfall.

Q_t is:

the dollar-valued orthogonal complement created when CAPM's risk premium is reinterpreted as a phase rotation. (6.20)

In plain English:

Q_t is the part of the base-discounted cash-flow amplitude that CAPM risk prevents from appearing as real admitted value. (6.21)

This gives Q a clear role.

Ordinary CAPM says:

risk reduces R_t. (6.22)

Phase-twisted CAPM says:

risk rotates A_t into R_t and Q_t. (6.23)

The first statement gives a scalar answer.

The second statement preserves the lost structure.

7. The Complex Cash-Flow State

7.1 One-period complex value

For each future cash flow, the phase-twisted model defines a complex cash-flow state:

$$Z_t = R_t + iQ_t. \quad (7.1)$$

Since:

$$R_t = A_t \cos \theta_t, \quad (7.2)$$

and:

$$Q_t = A_t \sin \theta_t, \quad (7.3)$$

we can also write:

$$Z_t = A_t \cos \theta_t + iA_t \sin \theta_t. \quad (7.4)$$

Factor out A_t:

$$Z_t = A_t(\cos \theta_t + i \sin \theta_t). \quad (7.5)$$

Using Euler's identity:

$\cos \theta_t + i \sin \theta_t = e^{(i\theta_t)}$, (7.6)

we get:

$Z_t = A_t e^{(i\theta_t)}$. (7.7)

So the one-period phase-twisted valuation is:

$Z_t = [CF_t/(1 + r_{base})^t] e^{(i\theta_t)}$. (7.8)

where:

$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t$. (7.9)

7.2 Multi-period complex DCF

For multiple future cash flows, define:

$Z = \Sigma_t Z_t$. (7.10)

Therefore:

$Z = \Sigma_t [CF_t/(1 + r_{base})^t] e^{(i\theta_t)}$. (7.11)

This is the complex DCF state.

Its real part is:

$R = Re(Z)$. (7.12)

Its imaginary part is:

$Q = Im(Z)$. (7.13)

So:

$Z = R + iQ$. (7.14)

This is the multi-period counterpart of:

$Z_t = R_t + iQ_t$. (7.15)

7.3 Important warning: summing Q_t is not always the same as taking total Q

In a one-period model:

$Q = Q_t$. (7.16)

In a multi-period model, if all Q_t terms point in the same imaginary direction, then:

$Q = \Sigma_t Q_t$. (7.17)

But if different periods have different phase angles, then the full complex sum matters:

$Z = \Sigma_t A_t e^{(i\theta_t)}$. (7.18)

Then:

$R = \Sigma_t A_t \cos \theta_t$. (7.19)

$Q = \Sigma_t A_t \sin \theta_t$. (7.20)

and:

$|Z|^2 = R^2 + Q^2$. (7.21)

This is not necessarily equal to:

$\Sigma_t A_t^2$. (7.22)

because different cash-flow vectors can interfere geometrically when added.

That is not a problem. It is actually useful.

It means multi-period valuation is not merely a pile of independent cash flows. It becomes a phase-weighted structure.

In ordinary DCF, all discounted cash flows are simply added on the real axis.

In phase-twisted DCF, each future cash flow has:

an amplitude A_t, (7.23)

a real projection R_t, (7.24)

an imaginary pressure Q_t, (7.25)

and a CAPM risk angle θ_t. (7.26)

7.4 Norm relation

Once the full complex value is written as:

$Z = R + iQ$, (7.27)

its magnitude is:

$|Z| = \sqrt{(R^2 + Q^2)}$. (7.28)

Therefore:

$|Z|^2 = R^2 + Q^2$. (7.29)

If the model is calibrated so that the observed market price P_0 equals the magnitude of Z, then:

$P_0 = |Z|$. (7.30)

and:

$P_0^2 = R^2 + Q^2$. (7.31)

This is the relationship we were searching for.

But it must be interpreted carefully.

$P_0^2 = R^2 + Q^2$ does **not** mean that standard CAPM already says market price is a Pythagorean norm.

It means:

if market price is interpreted as the magnitude of a complex valuation state, then the observed price can be decomposed into admitted CAPM value and market-implied orthogonal pressure. (7.32)

That gives a new diagnostic layer.

---

8. Formula and Framework Comparison Table

The following table is central to the article. It shows why Q is not merely decorative.

Framework	Main formula	What risk does	What is preserved	What is hidden	What Q adds
Standard CAPM	$r_{CAPM} = r_f + \beta ERP$	Converts systematic risk into required return	Clean expected-return discipline	No dollar-valued risk-pressure coordinate	Q is not needed for the Security Market Line itself
Risk-adjusted DCF	$PV = \Sigma CF_t/(1 + r_{CAPM})^t$	Reduces present value through higher r	Scalar CAPM-admitted value	The removed value-pressure disappears	Q preserves the hidden complement
Certainty-equivalent DCF	$PV = \Sigma CE_t/(1 + r_{base})^t$	Converts risky cash flow into safer equivalent cash flow	Risk-adjusted cash-flow interpretation	No explicit orthogonal pressure axis	Q reveals CE as real projection
Phase-twisted DCF	$A_t^2 = R_t^2 + Q_t^2$	βERP becomes angle θ_t	Real value and risk pressure both retained	More variables; interpretation required	CAPM becomes projection geometry
Market-calibrated complex value	$P_0^2 = R^2 + Q_{market}^2$	Market price implies residual pressure	CAPM value plus market-implied pressure	Requires convention when $P_0 < R$	Enables ΔQ diagnostic
Multi-factor extension	$A_t^2 = R_t^2 + \Sigma_k Q_{k,t}^2$	Each factor becomes a pressure dimension	Factor-risk decomposition in dollar geometry	Orthogonality must be tested	Generalizes Q beyond one-factor CAPM

The table shows a disciplined reason for adding Q.

Standard CAPM is already elegant as an expected-return model.

But when CAPM is inserted into DCF, it collapses value and risk into a single scalar.

Q restores the missing orthogonal coordinate.

So the contribution is not:

CAPM was incomplete because it lacked Q. (8.1)

The better statement is:

CAPM DCF hides a projection geometry that Q makes explicit. (8.2)

---

9. The Pressure Ratio: Beta as Real-to-Imaginary Tilt

9.1 Define the pressure ratio

The raw Q_t value has dollar units.

Sometimes a dimensionless ratio is more useful.

Define:

$q_t = Q_t/R_t$. (9.1)

This q_t is the **risk-pressure ratio**.

It measures how much imaginary pressure exists per unit of admitted real value.

Since:

$R_t = A_t \cos \theta_t$, (9.2)

and:

$Q_t = A_t \sin \theta_t$, (9.3)

we have:

$q_t = (A_t \sin \theta_t)/(A_t \cos \theta_t)$. (9.4)

Therefore:

$q_t = \tan \theta_t$. (9.5)

This is a very clean result.

The pressure ratio is simply the tangent of the CAPM risk angle.

9.2 Express q_t directly from CAPM variables

We know:

$\cos \theta_t = [(1 + r_{base})/(1 + r_{CAPM})]^t$. (9.6)

Since:

$\tan^2\theta_t = 1/\cos^2\theta_t - 1$, (9.7)

we get:

$q_t^2 = [(1 + r_{CAPM})/(1 + r_{base})]^{(2t)} - 1.$ (9.8)

So:

$q_t = \sqrt{\{[(1 + r_{CAPM})/(1 + r_{base})]^{(2t)} - 1\}}.$ (9.9)

Substitute:

$r_{CAPM} = r_{base} + \beta ERP.$ (9.10)

Then:

$q_t = \sqrt{\{[(1 + r_{base} + \beta ERP)/(1 + r_{base})]^{(2t)} - 1\}}.$ (9.11)

This is one of the strongest formulas in the article.

It shows beta as a value-tilt parameter.

9.3 Interpretation

In standard CAPM:

β measures systematic covariance exposure. (9.12)

In phase-twisted CAPM:

β controls the real-to-imaginary value tilt. (9.13)

The chain is:

$\beta \uparrow \rightarrow r_{CAPM} \uparrow \rightarrow \theta_t \uparrow \rightarrow q_t \uparrow.$ (9.14)

In words:

higher beta rotates more of the base-discounted cash-flow amplitude into Q. (9.15)

This is a useful teaching interpretation.

Beta is no longer only an abstract regression slope.

It becomes a geometric pressure generator.

9.4 Why q_t is useful

Two securities may have different cash-flow sizes, making Q_t hard to compare directly.

But $q_t = Q_t/R_t$ is scale-normalized.

It allows comparison across assets:

q_t small: most value remains real-admitted. (9.16)

q_t large: much value sits in market-risk pressure. (9.17)

For example:

a mature regulated utility may have small q_t; (9.18)

a long-duration growth stock may have large q_t; (9.19)

a story stock may have very high q_t. (9.20)

This creates a geometric classification of assets.

---

10. Inverse Beta from Q/R

10.1 Recover r_CAPM from q_t

Because:

$q_t = Q_t/R_t$, (10.1)

and:

$q_t^2 = [(1 + r_{CAPM})/(1 + r_{base})]^{(2t)} - 1$, (10.2)

we have:

$1 + q_t^2 = [(1 + r_{CAPM})/(1 + r_{base})]^{(2t)}$. (10.3)

Raise both sides to the power $1/(2t)$:

$(1 + q_t^2)^{(1/(2t))} = (1 + r_{CAPM})/(1 + r_{base})$. (10.4)

Therefore:

$1 + r_{CAPM} = (1 + r_{base})(1 + q_t^2)^{(1/(2t))}$. (10.5)

So:

$r_{CAPM} = (1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1$. (10.6)

This is the inverse discount-rate formula.

It says:

given q_t, we can recover the CAPM discount rate implied by the real/imaginary pressure ratio. (10.7)

10.2 Recover beta

Since:

$r_{CAPM} = r_{base} + \beta ERP$, (10.8)

we get:

$\beta ERP = r_{CAPM} - r_{base}$. (10.9)

Substitute the inverse formula:

$\beta ERP = [(1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1] - r_{base}$. (10.10)

Therefore:

$\beta = \{[(1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1] - r_{base}\}/ERP$. (10.11)

This is important.

It shows that beta can be re-expressed through the pressure ratio q_t.

So q_t is not arbitrary. It contains the same systematic-risk information, but in geometric form.

10.3 Interpretation

Standard CAPM obtains beta from covariance:

$\beta_i = Cov(R_i,R_m)/Var(R_m)$. (10.12)

Phase-twisted CAPM gives another interpretation:

β is the market-risk loading required to generate a given Q/R tilt. (10.13)

These are not competing definitions.

The covariance beta remains the ordinary empirical estimate.

The Q/R expression is a geometric reinterpretation of what that beta does inside valuation.

So:

covariance beta explains where systematic risk comes from; (10.14)

Q/R beta explains where systematic risk goes in DCF. (10.15)

That sentence is likely worth keeping in the final article.

---

11. CAPM Value Haircut as a Quadratic Pressure Law

11.1 Define the value haircut

The base-discounted amplitude is:

$A_t = CF_t/(1 + r_{base})^t$. (11.1)

The CAPM-admitted real value is:

$R_t = A_t \cos \theta_t$. (11.2)

The difference is:

$Loss_t = A_t - R_t$. (11.3)

Substitute R_t:

$$\text{Loss}_t = A_t - A_t \cos \theta_t. \quad (11.4)$$

So:

$$\text{Loss}_t = A_t(1 - \cos \theta_t). \quad (11.5)$$

This is the CAPM value haircut.

It is the amount by which CAPM risk rotation reduces the real-axis value relative to base-discounted amplitude.

11.2 Small-angle approximation

For small θ_t:

$$\cos \theta_t \approx 1 - \theta_t^2/2. \quad (11.6)$$

Therefore:

$$1 - \cos \theta_t \approx \theta_t^2/2. \quad (11.7)$$

So:

$$\text{Loss}_t \approx A_t\theta_t^2/2. \quad (11.8)$$

Also, for small θ_t:

$$\sin \theta_t \approx \theta_t. \quad (11.9)$$

Since:

$$Q_t = A_t \sin \theta_t, \quad (11.10)$$

we get:

$$Q_t \approx A_t\theta_t. \quad (11.11)$$

Therefore:

$$\theta_t \approx Q_t/A_t. \quad (11.12)$$

Substitute into the haircut approximation:

$$\text{Loss}_t \approx A_t(Q_t/A_t)^2/2. \quad (11.13)$$

So:

$$\text{Loss}_t \approx Q_t^2/(2A_t). \quad (11.14)$$

This is a very elegant relationship.

11.3 Interpretation

Ordinary CAPM says:

risk premium raises the discount rate and lowers present value. (11.15)

Phase-twisted CAPM says:

the value haircut is approximately quadratic in Q. (11.16)

In compact form:

CAPM haircut ≈ risk-pressure² / twice base amplitude. (11.17)

This makes the risk adjustment feel more physical.

Small Q produces a small value haircut.

Large Q produces a disproportionately larger haircut.

This may help explain why high-beta, long-duration cash flows can be extremely sensitive to changes in discount assumptions.

---

12. Certainty-Equivalent Valuation as a Real Projection

12.1 Ordinary certainty-equivalent form

Certainty-equivalent valuation separates risk adjustment from discounting.

Instead of discounting expected risky cash flow at a risky discount rate, it converts the cash flow into a certainty-equivalent amount, then discounts at a base or risk-free rate.

Write:

$R_t = CE_t/(1 + r_{base})^t$. (12.1)

Here CE_t is the certainty-equivalent cash flow.

12.2 Compare with phase-twisted form

In phase-twisted CAPM:

$R_t = A_t \cos \theta_t$. (12.2)

Since:

$A_t = CF_t/(1 + r_{base})^t$, (12.3)

we have:

$R_t = [CF_t/(1 + r_{base})^t] \cos \theta_t$. (12.4)

Rewrite:

$R_t = [CF_t \cos \theta_t]/(1 + r_{base})^t$. (12.5)

Compare with certainty-equivalent form:

$R_t = CE_t/(1 + r_{base})^t$. (12.6)

Therefore:

$CE_t = CF_t \cos \theta_t$. (12.7)

This is a strong bridge to mature valuation theory.

12.3 Interpretation

Certainty-equivalent cash flow is the real projection of risky cash flow. (12.8)

The imaginary complement is:

$CF_t \sin \theta_t$. (12.9)

After base discounting, that becomes:

$Q_t = [CF_t/(1 + r_{base})^t] \sin \theta_t$. (12.10)

So the certainty-equivalent view and the Q-view are naturally compatible.

Certainty-equivalent valuation says:

risk converts CF_t into CE_t. (12.11)

Phase-twisted valuation says:

risk projects CF_t into CE_t and an orthogonal pressure component. (12.12)

The certainty-equivalent method keeps the real projection.

The Q-framework keeps both projection and complement.

That is the advantage.

12.4 Why this matters

This section helps protect the article from looking like an arbitrary invention.

Q is not created out of nowhere.

It arises when we ask:

If certainty-equivalent cash flow is the real projection, what is the orthogonal complement? (12.13)

The answer is Q.

So Q can be understood as:

the discounted imaginary complement of certainty-equivalent cash-flow reduction. (12.14)

This is one of the strongest theoretical justifications for the framework.

13. Market-Implied Q and the ΔQ Diagnostic

13.1 Why market price creates a second Q

So far, Q has been derived from CAPM itself.

That gives:

Q_{CAPM}. (13.1)

But the market also gives us an observed price:

P_0. (13.2)

If we interpret market price as the magnitude of a complex valuation state, then:

$P_0^2 = R^2 + Q_{market}^2$. (13.3)

Therefore:

$Q_{market} = \sqrt{(P_0^2 - R^2)}$. (13.4)

This Q_{market} is not the same as Q_{CAPM}.

Q_{CAPM} is the risk pressure implied by the CAPM phase-twisted DCF model.

Q_{market} is the risk / option / residual pressure implied by the actual market price.

So we now have two Q-values:

Q_{CAPM} = model-implied risk pressure. (13.5)

Q_{market} = market-implied residual pressure. (13.6)

This creates a diagnostic.

13.2 Define ΔQ

Define:

$\Delta Q = Q_{market} - Q_{CAPM}$. (13.7)

This is the residual pressure gap.

It asks:

How much more or less pressure does the market contain compared with CAPM phase geometry? (13.8)

If:

$\Delta Q \approx 0$, (13.9)

then the market price is broadly consistent with CAPM phase-risk geometry.

If:

$\Delta Q > 0$, (13.10)

then the market price contains more orthogonal pressure than CAPM explains.

If:

$\Delta Q < 0$, (13.11)

then the market price contains less orthogonal pressure than CAPM predicts, or the CAPM DCF value R may be too high.

13.3 Interpretation table

Condition	Meaning
$\Delta Q \approx 0$	Market price is consistent with CAPM phase-risk geometry
$\Delta Q > 0$	Market contains extra growth optionality, narrative premium, liquidity premium, speculation, hidden demand, or non-CAPM priced upside
$\Delta Q < 0$	Market distrusts the cash flows, prices hidden distress, rejects the valuation model, or imposes a non-CAPM penalty
Q_market high, Q_CAPM low	Market is pricing something CAPM beta does not capture
Q_market low, Q_CAPM high	CAPM risk looks high, but market is not rewarding or recognizing the associated pressure component

This is potentially the most practical part of the framework.

Standard CAPM gives a required return.

Phase-twisted CAPM gives a model-implied pressure coordinate.

Market calibration gives a market-implied pressure coordinate.

The difference is:

ΔQ. (13.12)

13.4 Why ΔQ may be more informative than price gap

A normal valuation gap is:

$\text{Gap} = P_0 - R$. (13.13)

This is useful, but it mixes scale and geometry.

If a stock trades above CAPM DCF value, the ordinary interpretation is often:

overvalued. (13.14)

But the phase-twisted interpretation is more nuanced.

If:

$P_0 > R$, (13.15)

then perhaps the market is pricing:

growth options,

strategic optionality,

liquidity preference,

narrative momentum,

takeover probability,

network effects,

scarcity premium,

or investor disagreement.

These may not be captured by CAPM beta.

So instead of saying:

$P_0 - R$ = overvaluation, (13.16)

we say:

$P_0^2 - R^2$ = market-implied orthogonal pressure^2. (13.17)

This is more expressive.

It avoids forcing every price-above-DCF situation into the same category.

13.5 Signed convention when $P_0 < R$

The formula:

$Q_{market} = \sqrt{(P_0^2 - R^2)}$ (13.18)

requires:

$P_0 \geq R$. (13.19)

But markets often price securities below a model's estimated value.

When:

$P_0 < R$, (13.20)

we need a convention.

There are at least three possible conventions.

Convention A — no real Q, report negative gap

If $P_0 < R$, do not compute real Q_{market}.

Instead report:

$DiscountGap = R - P_0$. (13.21)

This is conservative.

Convention B — signed imaginary pressure

Define:

$Q_{market} = \text{sign}(P_0 - R)\sqrt{|P_0^2 - R^2|}$. (13.22)

Then Q_{market} can be negative.

Positive Q means market-added pressure.

Negative Q means market-subtracted pressure.

Convention C — distress axis

Use a second imaginary direction:

$Z = R + iQ_{up} - iQ_{down}$. (13.23)

Here:

Q_{up} = market-implied upside / optionality pressure. (13.24)

Q_{down} = market-implied distrust / distress pressure. (13.25)

This may be useful in a more advanced version.

For the first article, Convention B is probably simplest:

$Q_{market} = \text{sign}(P_0 - R)\sqrt{|P_0^2 - R^2|}$. (13.26)

But for introductory clarity, the article can first focus on $P_0 \geq R$.

---

14. Duration of Risk Pressure

14.1 Why Q is naturally horizon-sensitive

The pressure ratio is:

$q_t = Q_t/R_t$. (14.1)

We derived:

$q_t = \sqrt{[(1 + r_{CAPM})/(1 + r_{base})]^{(2t)} - 1}$. (14.2)

This expression grows with t when $r_{CAPM} > r_{base}$.

That means longer-dated cash flows naturally carry larger Q relative to R.

In plain English:

the farther away the cash flow, the more CAPM risk rotation accumulates. (14.3)

This matches an important finance intuition.

Long-duration equities, especially growth stocks, are more sensitive to discount-rate and risk-premium changes.

The Q-framework makes this visible:

long-duration value carries more imaginary market-risk pressure. (14.4)

14.2 Define R-duration

Ordinary duration measures the timing of value.

In this framework, define R-duration:

$D_R = \Sigma_t tR_t / \Sigma_t R_t$. (14.5)

This is the value-weighted average timing of real admitted value.

If D_R is high, most real value comes from distant cash flows.

If D_R is low, most real value comes from near-term cash flows.

14.3 Define Q-duration

Now define Q-duration:

$D_Q = \Sigma_t tQ_t / \Sigma_t Q_t$. (14.6)

This measures where the risk pressure is concentrated in time.

If D_Q is high, the imaginary pressure is concentrated in distant future cash flows.

If D_Q is low, the pressure sits mostly in near-term uncertainty.

This is a new diagnostic.

Standard DCF often asks:

Where is the value? (14.7)

The Q-framework also asks:

Where is the pressure? (14.8)

14.4 Compare D_Q and D_R

The difference:

$D_Q - D_R$ (14.9)

may be very informative.

If:

$D_Q > D_R$, (14.10)

then risk pressure is more long-dated than admitted value.

This is typical of assets where near-term cash flows are visible but most upside or uncertainty lies in the distant future.

If:

$D_Q \approx D_R$, (14.11)

then risk pressure and admitted value are distributed similarly across time.

If:

$D_Q < D_R$, (14.12)

then near-term risk pressure dominates relative to long-term value.

That may occur in restructuring, litigation, regulatory review, refinancing risk, or short-term crisis situations.

14.5 Growth stocks in R–Q duration space

Growth stocks often have:

low near-term R,

large distant A_t,

large θ_t for distant t,

and therefore large distant Q_t.

So:

D_Q high. (14.13)

This gives a geometric interpretation of growth-stock sensitivity.

When discount rates rise, or ERP rises, the angle θ_t increases.

For long t, even small increases in θ_t can create large movement between R_t and Q_t.

The result is:

growth-stock valuation is not merely high; it is phase-sensitive. (14.14)

That sentence may be useful in the final article.

---

15. Security Market Line as an Angle Curve

15.1 Standard Security Market Line

The standard Security Market Line, or SML, is:

$E[R_i] = r_f + \beta_i ERP$. (15.1)

This is linear in beta.

If β_i rises by one unit, the expected return rises by one equity risk premium.

It is one of the cleanest parts of CAPM.

The Q-framework does not simplify this equation.

It translates it into an angle geometry.

15.2 Phase form of the Security Market Line

Using the phase definition:

$\cos \theta_t(\beta_i) = [(1 + r_{base})/(1 + r_{base} + \beta_{iERP})]^t$. (15.2)

Therefore:

$\theta_t(\beta_i) = \arccos\{[(1 + r_{base})/(1 + r_{base} + \beta_{iERP})]^t\}$. (15.3)

This is the Security Market Line rewritten as a risk-angle curve.

In the ordinary SML:

$\beta_i \rightarrow$ expected return. (15.4)

In phase-CAPM:

$\beta_i \rightarrow$ risk angle. (15.5)

Then:

risk angle → R/Q split. (15.6)

So the path becomes:

$\beta_i \rightarrow \theta_t \rightarrow q_t \rightarrow R_t$ and Q_t. (15.7)

15.3 Small-risk approximation

From Section 4:

$\theta_t \approx \sqrt{[2t\beta_{iERP}/(1 + r_{base})]}$. (15.8)

Square both sides:

$\theta_t^2 \approx 2t\beta_{iERP}/(1 + r_{base})$. (15.9)

Therefore:

$\beta_i \approx [(1 + r_{base})\theta_t^2]/(2tERP)$. (15.10)

This means:

CAPM is linear in beta, but approximately quadratic in risk angle. (15.11)

Equivalently:

beta is proportional to squared phase displacement. (15.12)

This is conceptually elegant.

It suggests that the SML may be read as a straight line only after the angle geometry has been squared and compressed.

15.4 What this adds

The standard SML answers:

What expected return compensates beta risk? (15.13)

The phase SML answers:

What valuation angle does beta risk impose on each future cash-flow horizon? (15.14)

Those are different questions.

The first belongs to asset pricing.

The second belongs to valuation geometry.

So Q and θ do not replace the SML.

They provide an extra lens for what the SML does when used inside DCF.

---

16. Alpha as Angular or Q-Residual Mispricing

16.1 Standard alpha

Standard CAPM alpha is:

$\alpha_i = E[R_i] - [r_f + \beta_i ERP]$. (16.1)

Alpha measures excess return relative to CAPM expectation.

If:

$\alpha_i > 0$, (16.2)

the asset outperforms CAPM prediction.

If:

$\alpha_i < 0$, (16.3)

the asset underperforms CAPM prediction.

This is a return-space residual.

16.2 Q-residual alpha

In the Q-framework, we can define a pressure-space residual:

$\alpha_Q = Q_{market} - Q_{CAPM}$. (16.4)

This is not the same as ordinary alpha.

Ordinary alpha asks:

Is the return higher or lower than CAPM predicts? (16.5)

Q-alpha asks:

Is the market carrying more or less orthogonal pressure than CAPM phase geometry predicts? (16.6)

This can detect something different.

A stock may have zero historical alpha but still have a large Q_{market} gap if investors are pricing long-term optionality, narrative premium, or hidden risk.

16.3 Angular alpha

We may also define an angular residual.

Let:

θ_{CAPM} = CAPM-implied risk angle. (16.7)

Let:

θ_{market} = market-implied angle. (16.8)

Then:

$\alpha_\theta = \theta_{market} - \theta_{CAPM}$. (16.9)

This gives a dimensionless structural residual.

If:

$\alpha_\theta > 0$, (16.10)

the market-implied valuation state is more rotated than CAPM predicts.

If:

$\alpha_\theta < 0$, (16.11)

the market-implied valuation state is less rotated than CAPM predicts.

16.4 Three kinds of residual

We now have three different residuals:

Residual	Formula	Domain	Meaning
Price gap	$P_0 - R$	dollar price space	market price above or below model value

Residual	Formula	Domain	Meaning
Return alpha	$\alpha_i = E[R_i] - r_{CAPM}$	expected-return space	return unexplained by beta
Q-alpha	$\alpha_Q = Q_{market} - Q_{CAPM}$	pressure space	orthogonal market pressure unexplained by CAPM
Angle alpha	$\alpha_\theta = \theta_{market} - \theta_{CAPM}$	geometry space	structural rotation mismatch

This table is useful because it shows why Q is not just another name for alpha.

Alpha is a return residual.

Q-alpha is a pressure residual.

Angle-alpha is a geometry residual.

They may correlate, but they are not the same concept.

---

17. Value, Growth, and Story Stocks in R–Q Space

17.1 Why R alone is not enough

Traditional valuation often compares price to intrinsic value:

P_0 versus R. (17.1)

If:

$P_0 > R$, (17.2)

the stock may be called expensive or overvalued.

But this may be too crude.

A stock may trade above R because the market is pricing Q-like components:

growth optionality,

future platform value,

network effects,

strategic scarcity,

regulatory license value,

takeover probability,

or narrative-driven demand.

Some of these may later become real cash flows.

Some may disappear.

The point is not that Q is always “good.”

The point is that Q distinguishes:

price above R because of structured optionality, (17.3)

from:

price above R because of pure overpayment. (17.4)

That distinction is useful.

17.2 Classification by R/P0 and Q/P0

If:

$P_0^2 = R^2 + Q^2$, (17.5)

then:

$R/P_0 = \cos\theta$. (17.6)

and:

$Q/P_0 = \sin\theta$. (17.7)

This gives a clean classification.

Stock type	R/P0	Q/P0	θ	Interpretation
Bond-like equity	Very high	Very low	Small	Most price is admitted cash-flow value
Mature utility	High	Low/ moderate	Small	Stable value with limited pressure
Cyclical stock	Moderate	Moderate	Variable	Value and pressure shift with cycle
Growth stock	Lower	Higher	Large	Much price depends on future optionality
Story stock	Low	Very high	Very large	Price dominated by unresolved narrative pressure
Distress stock	unstable	high or negative	unstable	Market carries distrust, crisis pressure, or collapse risk

This table may become one of the most reader-friendly sections of the article.

17.3 Value stock interpretation

A value stock in this geometry has:

R/P0 high. (17.8)

Q/P0 low. (17.9)

θ small. (17.10)

This means most of the market price can be explained by admitted cash-flow value.

There is little additional unresolved pressure.

Such an asset may be easier to value with ordinary DCF.

17.4 Growth stock interpretation

A growth stock has:

R/P_0 lower. (17.11)

Q/P_0 higher. (17.12)

θ larger. (17.13)

This does not automatically mean the stock is bad.

It means more of the price depends on future unresolved structure.

The market is paying for:

long-dated cash flows,

scalable opportunity,

optionality,

strategic positioning,

or future regime change.

The Q-framework makes this explicit.

Growth stock = high-Q valuation state. (17.14)

17.5 Story stock interpretation

A story stock has:

R/P_0 very low. (17.15)

Q/P_0 very high. (17.16)

θ very large. (17.17)

This means the market price is dominated by unresolved pressure.

That pressure may be:

real future optionality,

speculative crowding,

narrative momentum,

liquidity imbalance,

or collective imagination.

The framework does not decide automatically whether the Q is justified.

It only says:

the price is not mainly sitting on the real admitted cash-flow axis. (17.18)

That is already a useful diagnostic.

---

18. Multi-Factor Extension: From One Q to Many Qs

18.1 Beyond CAPM

CAPM is a one-factor model.

It uses market beta as the central priced risk loading.

But mature finance often extends beyond CAPM using factor models.

A general factor discount rate may be written:

$r = r_base + \beta_1\lambda_1 + \beta_2\lambda_2 + \ldots + \beta_n\lambda_n$. (18.1)

where:

β_k = exposure to factor k. (18.2)

λ_k = risk premium for factor k. (18.3)

Examples may include:

market factor,

size factor,

value factor,

momentum factor,

quality factor,

liquidity factor,

term factor,

credit factor.

The Q-framework can be extended.

18.2 Single-angle multi-factor version

First define the total factor premium:

$\Lambda = \beta_1\lambda_1 + \beta_2\lambda_2 + \ldots + \beta_n\lambda_n$. (18.4)

Then:

$r = r_{base} + \Lambda$. (18.5)

The phase angle becomes:

$\cos \theta_t = [(1 + r_{base})/(1 + r_{base} + \Lambda)]^t$. (18.6)

Then:

$Q_t = A_t \sqrt{1 - [(1 + r_{base})/(1 + r_{base} + \Lambda)]^{2t}}$. (18.7)

This is the simplest extension.

All factor premia combine into one total risk angle.

This preserves the CAPM-style structure.

18.3 Multi-axis pressure version

A more ambitious extension assigns each factor its own pressure coordinate:

$A_t^2 = R_t^2 + Q_{1,t}^2 + Q_{2,t}^2 + \ldots + Q_{n,t}^2$. (18.8)

Here:

$Q_{1,t}$ = market-risk pressure. (18.9)

$Q_{2,t}$ = size-risk pressure. (18.10)

$Q_{3,t}$ = value-risk pressure. (18.11)

$Q_{4,t}$ = momentum-risk pressure. (18.12)

and so on.

This gives a multi-dimensional valuation geometry.

The complex plane becomes a higher-dimensional value-pressure space.

18.4 Orthogonality problem

The multi-axis version is attractive, but it requires caution.

Factor pressures are not automatically orthogonal.

Market, value, size, momentum, quality, and liquidity factors may overlap.

So the equation:

$A_t^2 = R_t^2 + \Sigma_k Q_{k,t}^2$ (18.13)

requires either:

orthogonalized factors,

principal components,

Gram-Schmidt-like transformation,

or covariance-adjusted metric.

A safer general version is:

$A_t^2 = R_t^2 + \vec{Q}_t^T G \vec{Q}_t$. (18.14)

where G is a factor-pressure metric.

If G is identity, the factors are orthogonal.

If G is not identity, the geometry accounts for overlap.

This is probably the correct mature extension.

18.5 Why the extension matters

The single Q version says:

CAPM risk pressure is not lost; it becomes Q. (18.15)

The multi-Q version says:

factor premia are not merely added into discount rates; they can be decomposed into pressure directions. (18.16)

This could become a broader framework:

Factor model = pressure geometry under a declared metric. (18.17)

This is not necessary for the first article, but it is an important future direction.

It may also become a bridge between:

CAPM,

Fama-French,

APT,

principal-component risk models,

and complex valuation geometry.

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19. Practical Worked Example

19.1 Setup

Consider a one-period expected cash flow:

$CF_1 = 110$. (19.1)

Let:

$r_{base} = 3\%$. (19.2)

$ERP = 5\%$. (19.3)

$\beta = 1.2$. (19.4)

Then:

$r_{CAPM} = r_{base} + \beta ERP$. (19.5)

So:

$r_{CAPM} = 3\% + 1.2 \times 5\%$. (19.6)

Therefore:

$r_{CAPM} = 9\%$. (19.7)

19.2 Compute base amplitude

The base-discounted amplitude is:

$A_1 = CF_1/(1 + r_{base})$. (19.8)

Substitute:

$A_1 = 110/1.03$. (19.9)

So:

$A_1 \approx 106.80$. (19.10)

19.3 Compute CAPM real value

The ordinary CAPM DCF value is:

$R_1 = CF_1/(1 + r_{CAPM})$. (19.11)

Substitute:

$R_1 = 110/1.09$. (19.12)

So:

$R_1 \approx 100.92$. (19.13)

This is the ordinary scalar CAPM value.

19.4 Compute Q

Using:

$Q_1 = \sqrt{(A_1^2 - R_1^2)}$, (19.14)

we get:

$Q_1 = \sqrt{(106.80^2 - 100.92^2)}$. (19.15)

Compute approximate values:

$106.80^2 \approx 11406.24$. (19.16)

$100.92^2 \approx 10184.85$. (19.17)

So:

$Q_1 \approx \sqrt{1221.39}$. (19.18)

Therefore:

$Q_1 \approx 34.95$. (19.19)

So the phase-twisted state is:

$Z_1 \approx 100.92 + i34.95$. (19.20)

and:

$|Z_1| \approx 106.80$. (19.21)

19.5 Compute pressure ratio

The pressure ratio is:

$q_1 = Q_1/R_1$. (19.22)

So:

$q_1 \approx 34.95/100.92$. (19.23)

Therefore:

$q_1 \approx 0.346$. (19.24)

This means:

the CAPM-implied imaginary pressure is about 34.6% of the real admitted value. (19.25)

19.6 Compute risk angle

The angle satisfies:

$\cos \theta_1 = R_1/A_1$. (19.26)

So:

$\cos \theta_1 \approx 100.92/106.80$. (19.27)

Therefore:

$\cos \theta_1 \approx 0.945$. (19.28)

Thus:

$\theta_1 \approx \arccos(0.945)$. (19.29)

So:

$\theta_1 \approx 19.1^\circ$. (19.30)

Interpretation:

The CAPM beta premium rotates the base-discounted cash-flow amplitude by about 19.1 degrees away from the real axis. (19.31)

19.7 Add market price

Suppose the observed market price is:

$P_0 = 105$. (19.32)

Then market-implied Q is:

$Q_{market} = \sqrt{(P_0^2 - R_1^2)}$. (19.33)

Substitute:

$Q_{market} = \sqrt{(105^2 - 100.92^2)}$. (19.34)

$105^2 = 11025$. (19.35)

$100.92^2 \approx 10184.85$. (19.36)

So:

$Q_{market} \approx \sqrt{840.15}$. (19.37)

Therefore:

$Q_{market} \approx 28.99$. (19.38)

The CAPM-implied Q was:

$Q_{CAPM} \approx 34.95$. (19.39)

Therefore:

$\Delta Q = Q_{market} - Q_{CAPM}$. (19.40)

So:

$\Delta Q \approx 28.99 - 34.95$. (19.41)

Thus:

$\Delta Q \approx -5.96$. (19.42)

Interpretation:

The market price implies less orthogonal pressure than the CAPM phase geometry predicts. (19.43)

Possible explanations:

market distrusts the cash-flow estimate,

the CAPM beta is too high,

the ERP input is too high,

the market is applying a different baseline,

or the observed price is discounting non-CAPM risk not captured in this one-period example.

19.8 What the example teaches

The ordinary CAPM DCF gives:

$R_1 \approx 100.92$. (19.44)

The phase-twisted CAPM gives:

$Z_1 \approx 100.92 + i34.95$. (19.45)

The market price gives:

$Q_{market} \approx 28.99$. (19.46)

The pressure gap gives:

$\Delta Q \approx -5.96$. (19.47)

So the new framework does not merely say:

value is 100.92. (19.48)

It says:

real admitted value is 100.92;

CAPM-implied risk pressure is 34.95;

market-implied pressure is 28.99;

pressure gap is −5.96. (19.49)

That is a richer diagnostic picture.

20. Limitations and Falsifiability

20.1 Q is not standard CAPM

The first limitation must be stated clearly:

Q is not a standard CAPM variable. (20.1)

Standard CAPM has:

β,

r_f,

r_m,

ERP,

r_CAPM,

expected return,

and market covariance.

It does not contain an imaginary-axis dollar pressure coordinate.

Therefore this article should not say:

CAPM already has Q. (20.2)

The correct statement is:

Q is a CAPM-derived coordinate introduced by reinterpreting risk-adjusted discounting as projection geometry. (20.3)

That distinction is important.

20.2 Q is not a free variable

Although Q is new, it is not arbitrary.

Given:

CF_t,

r_base,

β,

ERP,

t,

we define:

$$Q_t = [CF_t/(1 + r_{base})^t] \cdot \sqrt{1 - [(1 + r_{base})/(1 + r_{base} + \beta ERP)]^{(2t)}}. \quad (20.4)$$

So in the CAPM-implied version, Q_t is fully determined.

It is not an extra adjustable risk number.

This protects the framework from becoming loose metaphor.

The discipline is:

CAPM inputs determine θ_t; θ_t determines Q_t. (20.5)

20.3 Orthogonality is a modeling choice

The equation:

$A_t^2 = R_t^2 + Q_t^2$ (20.6)

uses Euclidean orthogonality.

That is mathematically clean, but it is also a modeling choice.

It assumes that admitted value and retained risk pressure can be treated as perpendicular components of one valuation amplitude.

This is reasonable as a first geometric representation, but it should not be overstated.

A more general model may need:

$A_t^2 = R_t^2 + gQ_t^2$, (20.7)

where g is a metric coefficient.

Or, for multiple factors:

$A_t^2 = R_t^2 + \vec{Q}_t^T G \vec{Q}_t$. (20.8)

where G is a factor-pressure metric.

So the simple model uses:

$G = I$. (20.9)

That is the identity metric.

Future empirical work may need a non-identity metric.

20.4 Multi-period aggregation can create phase effects

For one cash flow:

$A_t^2 = R_t^2 + Q_t^2$. (20.10)

For multiple cash flows:

$Z = \Sigma_t A_t e^{(i\theta_t)}$. (20.11)

Then:

$R = Re(Z)$. (20.12)

$Q = Im(Z)$. (20.13)

and:

$|Z|^2 = R^2 + Q^2$. (20.14)

But:

$|\Sigma_t A_t e^{(i\theta_t)}|^2 \neq \Sigma_t A_t^2$ in general. (20.15)

This means cash-flow components may interact through their phase structure.

For a first article, that can be treated simply:

Each cash flow has its own phase-twisted component. (20.16)

The full valuation is the sum of those components. (20.17)

But future articles may explore whether this phase interaction is useful for explaining valuation regimes, duration effects, or cash-flow uncertainty structures.

20.5 Market price may be below R

The formula:

$Q_{market} = \sqrt{(P_0^2 - R^2)}$ (20.18)

is real-valued only when:

$P_0 \geq R$. (20.19)

But in real markets, price can be below model value.

Therefore the article should define a convention.

A simple convention is:

$Q_{market} = \text{sign}(P_0 - R)\sqrt{|P_0^2 - R^2|}$. (20.20)

This allows positive and negative pressure.

Positive Q_market means:

market price contains additional orthogonal pressure above R. (20.21)

Negative Q_market means:

market price reflects distrust, distress, discount, or negative residual pressure relative to R. (20.22)

This signed convention is not standard finance.

It is a modeling convention.

The article should make that explicit.

20.6 Empirical falsifiability

The framework becomes more than a teaching geometry only if Q or ΔQ predicts, explains, or organizes something better than ordinary measures.

Possible tests include:

1. **Future returns**

Does ΔQ predict abnormal returns after controlling for beta, size, value, momentum, and quality?

2. **Forecast revisions**

Does high positive ΔQ predict later analyst upgrades, revenue surprises, or growth revisions?

3. **Drawdown sensitivity**

Do high-Q assets suffer larger drawdowns when ERP rises?

4. **Growth-stock duration**

Does D_Q explain interest-rate sensitivity better than ordinary equity duration?

5. **Option market relation**

Does Q_market correlate with implied volatility, skew, or option-implied uncertainty?

6. **Liquidity and narrative premium**

Does positive ΔQ correlate with liquidity demand, retail attention, news intensity, or social-media narrative strength?

7. **Distress detection**

Does negative Q_market or negative ΔQ predict credit stress, refinancing risk, or earnings disappointment?

If none of these tests works, then Q remains a teaching coordinate.

If some work, Q becomes a candidate empirical diagnostic.

20.7 The strongest falsifiable hypothesis

A clean falsifiable hypothesis would be:

Assets with high positive ΔQ contain market-implied residual pressure not explained by CAPM beta, and this residual pressure should correlate with future option-implied volatility, forecast dispersion, narrative intensity, or abnormal return reversal. (20.23)

A second hypothesis:

Assets with high D_Q are more sensitive to ERP and discount-rate shocks than assets with similar R-duration but lower Q-duration. (20.24)

These are testable.

They could be examined using:

market prices,

analyst cash-flow forecasts,

estimated beta,

ERP assumptions,

option-implied volatility,

forecast dispersion,

and realized future returns.

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21. Conclusion: Why Q May Be Worth Adding

21.1 CAPM remains elegant

CAPM remains elegant because it gives a simple expected-return relation:

$r_{CAPM} = r_f + \beta ERP$. (21.1)

This article does not improve that formula.

It does not make the Security Market Line simpler.

It does not replace beta.

It does not claim that Q is a missing official CAPM variable.

The article's claim is narrower:

When CAPM is used inside DCF, it hides a projection geometry. (21.2)

21.2 What scalar CAPM DCF hides

Risk-adjusted DCF gives:

$R_t = CF_t/(1 + r_{CAPM})^t$. (21.3)

This is useful, but it compresses the process.

It does not preserve the difference between:

base-discounted amplitude A_t, (21.4)

real admitted CAPM value R_t, (21.5)

and retained market-risk pressure Q_t. (21.6)

The phase-twisted model restores this structure:

$A_t^2 = R_t^2 + Q_t^2$. (21.7)

Now the value does not simply "shrink."

It rotates.

21.3 What Q adds

Q adds five things.

First, it preserves the hidden complement of CAPM discounting:

$Q_t = \sqrt{(A_t^2 - R_t^2)}$. (21.8)

Second, it converts beta premium into a pressure ratio:

$Q_t/R_t = \sqrt{\{[(1 + r_{base} + \beta ERP)/(1 + r_{base})]^{(2t)} - 1\}}$. (21.9)

Third, it gives an inverse route back to beta:

$\beta = \{[(1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1] - r_{base}\}/ERP$. (21.10)

Fourth, it turns the CAPM value haircut into an approximate quadratic pressure law:

$Loss_t \approx Q_t^2/(2A_t)$. (21.11)

Fifth, it creates a market diagnostic:

$\Delta Q = Q_{market} - Q_{CAPM}$. (21.12)

These relationships are the reason Q may be worth keeping.

21.4 The final interpretation

The phase-twisted reading can be summarized as:

CAPM risk premium is not only a discount-rate increment. It can also be interpreted as a phase angle that rotates cash-flow amplitude away from the real value axis. (21.13)

Then:

R is the admitted real projection. (21.14)

Q is the retained imaginary pressure. (21.15)

P_0, if interpreted as market magnitude, can be decomposed into:

$P_0^2 = R^2 + Q_{market}^2$. (21.16)

This gives a new way to read market price:

Market price may contain both admitted value and unresolved pressure. (21.17)

21.5 Closing line

Standard CAPM prices risk.

Phase-twisted CAPM remembers where the risk went. (21.18)

That is the conceptual advantage of Q.

Appendix A — Core Formula Summary

A.1 Standard CAPM

ERP = r_m − r_f. (A.1)

r_CAPM = r_base + βERP. (A.2)

If r_base = r_f:

r_CAPM = r_f + β(r_m − r_f). (A.3)

A.2 Base amplitude

A_t = CF_t/(1 + r_base)^t. (A.4)

A.3 CAPM risk angle

cos θ_t = [(1 + r_base)/(1 + r_CAPM)]^t. (A.5)

cos θ_t = [(1 + r_base)/(1 + r_base + βERP)]^t. (A.6)

θ_t = arccos{[(1 + r_base)/(1 + r_base + βERP)]^t}. (A.7)

A.4 Real projection

R_t = A_t cos θ_t. (A.8)

R_t = CF_t/(1 + r_CAPM)^t. (A.9)

R = Σ_t R_t. (A.10)

A.5 Imaginary pressure

Q_t = A_t sin θ_t. (A.11)

Q_t = [CF_t/(1 + r_base)^t] · √{1 − [(1 + r_base)/(1 + r_base + βERP)]^(2t)}. (A.12)

Q = Σ_t Q_t, if all components are aggregated on the same imaginary axis. (A.13)

More generally:

Z = Σ_t A_t e^(iθ_t). (A.14)

R = Re(Z). (A.15)

Q = Im(Z). (A.16)

A.6 Pythagorean identity

$A_t^2 = R_t^2 + Q_t^2$. (A.17)

$|Z|^2 = R^2 + Q^2$. (A.18)

If calibrated to market price:

$P_0 = |Z|$. (A.19)

Then:

$P_0^2 = R^2 + Q^2$. (A.20)

A.7 Pressure ratio

$q_t = Q_t/R_t$. (A.21)

$q_t = \tan \theta_t$. (A.22)

$q_t = \sqrt{[(1 + r_{CAPM})/(1 + r_{base})]^{(2t)} - 1}$. (A.23)

$q_t = \sqrt{[(1 + r_{base} + \beta ERP)/(1 + r_{base})]^{(2t)} - 1}$. (A.24)

A.8 Inverse beta

$r_{CAPM} = (1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1$. (A.25)

$\beta = \{[(1 + r_{base})(1 + q_t^2)^{(1/(2t))} - 1] - r_{base}\}/ERP$. (A.26)

A.9 Value haircut

$Loss_t = A_t - R_t$. (A.27)

$Loss_t = A_t(1 - \cos \theta_t)$. (A.28)

For small θ_t:

$Loss_t \approx A_t\theta_t^2/2$. (A.29)

Since $Q_t \approx A_t\theta_t$:

$Loss_t \approx Q_t^2/(2A_t)$. (A.30)

A.10 Certainty-equivalent bridge

$R_t = CE_t/(1 + r_{base})^t$. (A.31)

$R_t = [CF_t \cos \theta_t]/(1 + r_{base})^t$. (A.32)

Therefore:

$CE_t = CF_t \cos \theta_t$. (A.33)

The imaginary complement before base discounting is:

$CF_t \sin \theta_t$. (A.34)

The discounted imaginary pressure is:

$Q_t = [CF_t/(1 + r_{base})^t] \sin \theta_t$. (A.35)

A.11 Market-implied Q

If $P_0 \geq R$:

$Q_{market} = \sqrt{(P_0^2 - R^2)}$. (A.36)

Signed version:

$Q_{market} = \text{sign}(P_0 - R)\sqrt{|P_0^2 - R^2|}$. (A.37)

Pressure gap:

$\Delta Q = Q_{market} - Q_{CAPM}$. (A.38)

A.12 Q-duration

$D_R = \Sigma_t tR_t/\Sigma_t R_t$. (A.39)

$D_Q = \Sigma_t tQ_t/\Sigma_t Q_t$. (A.40)

If:

$D_Q > D_R$, (A.41)

risk pressure is more long-dated than admitted value.

Appendix B — Formula / Framework Comparison Table

Framework	Main object	Formula	Risk treatment	What becomes visible	What remains hidden	Role of Q
CAPM expected return	r_{CAPM}	$r_{CAPM} = r_f + \beta ERP$	Risk becomes required return	Systematic risk premium	Dollar value of risk pressure	Q not necessary
Risk-adjusted DCF	R	$R = \Sigma\, CF_t/(1 + r_{CAPM})^t$	Risk raises discount rate	Scalar present value	Removed value-pressure	Q recovers the hidden complement
Base-discounted amplitude	A_t	$A_t = CF_t/(1 + r_{base})^t$	Risk not yet applied	Pre-CAPM amplitude	No risk split yet	Q arises after rotation
Risk-angle form	θ_t	$\cos \theta_t = [(1 + r_{base})/(1 +$	Risk becomes angle	Phase effect of βERP	Dollar pressure not yet explicit	θ_t generates Q

Framework	Main object	Formula	Risk treatment	What becomes visible	What remains hidden	Role of Q
		r_CAPM)]^t				
Phase-twisted DCF	Z_t	$Z_t = A_t e^{(i\theta_t)}$	Risk rotates amplitude	R_t and Q_t	Requires interpretation	Q is imaginary projection
Pythagorean value split	A_t^2	$A_t^2 = R_t^2 + Q_t^2$	Risk becomes orthogonal complement	Value-pressure geometry	Assumes Euclidean metric	Q is preserved complement
Certainty-equivalent DCF	CE_t	$CE_t = CF_t \cos \theta_t$	Risk reduces cash flow	Real equivalent cash flow	Orthogonal pressure side	Q is the missing sine component
Market-implied complex value	P_0	$P_0^2 = R^2 + Q_{market}^2$	Price implies pressure	Market residual	Needs sign convention	Q_{market} measures priced pressure
Residual pressure diagnostic	ΔQ	$\Delta Q = Q_{market} - Q_{CAPM}$	Compares market vs model pressure	Non-CAPM residual	Cause must be interpreted	Q becomes diagnostic
Multi-factor pressure geometry	$\vec{Q}$	$A_t^2 = R_t^2 + \vec{Q}_t^T G \vec{Q}_t$	Factors become pressure directions	Factor-risk geometry	Metric must be estimated	Q generalizes to vector pressure

This table gives the article its strongest defense.

The new variable Q is justified because it creates relationships that existing scalar CAPM DCF does not show directly.

Appendix C — Suggested Visual Diagrams

C.1 Diagram 1 — Ordinary CAPM DCF

Flow:

$CF_t \rightarrow$ discount by $r_{CAPM} \rightarrow R_t$. (C.1)

Caption:

Standard CAPM DCF collapses risk-adjusted value into one scalar. (C.2)

C.2 Diagram 2 — Phase-twisted DCF

Flow:

$CF_t \rightarrow$ base discount $\rightarrow A_t \rightarrow$ CAPM phase rotation $\rightarrow R_t + iQ_t$. (C.3)

Caption:

The phase-twisted model preserves both admitted value and retained risk pressure. (C.4)

C.3 Diagram 3 — Right triangle

Axes:

horizontal axis = R_t, CAPM-admitted real value. (C.5)

vertical axis = Q_t, retained market-risk pressure. (C.6)

hypotenuse = A_t, base-discounted amplitude. (C.7)

angle = θ_t, CAPM risk angle. (C.8)

Key formula:

$A_t^2 = R_t^2 + Q_t^2$. (C.9)

C.4 Diagram 4 — Market-implied Q

Show:

R as horizontal base,

P_0 as market magnitude,

Q_{market} as vertical component.

Formula:

$Q_{market} = \sqrt{(P_0^2 - R^2)}$. (C.10)

Caption:

Market price can be read as a magnitude containing admitted value plus residual pressure. (C.11)

C.5 Diagram 5 — Q-duration

Show cash-flow bars over time with two curves:

R_t curve,

Q_t curve.

Highlight:

D_R = center of admitted value. (C.12)

D_Q = center of risk pressure. (C.13)

Caption:

Growth-like assets often have $D_Q > D_R$. (C.14)

Appendix D — Possible Empirical Research Protocol

D.1 Dataset

For each stock i and time τ, collect:

$P_{0,i,\tau}$ = market price. (D.1)

CF forecast path = analyst or model cash-flow estimates. (D.2)

$\beta_{i,\tau}$ = estimated CAPM beta. (D.3)

ERP_τ = equity risk premium estimate. (D.4)

$r_{base,\tau}$ = risk-free or declared base rate. (D.5)

option-implied volatility, if available. (D.6)

forecast dispersion. (D.7)

realized future return. (D.8)

liquidity measures. (D.9)

narrative or attention proxies. (D.10)

D.2 Compute

For each stock:

$A_t = CF_t/(1 + r_{base})^t$. (D.11)

$R_t = CF_t/(1 + r_{CAPM})^t$. (D.12)

$Q_{CAPM,t} = \sqrt{(A_t^2 - R_t^2)}$. (D.13)

$R = \Sigma_t R_t$. (D.14)

$Q_{CAPM} = \Sigma_t Q_{CAPM,t}$, or $Q_{CAPM} = Im(\Sigma_t A_t e^{(i\theta_t)})$. (D.15)

$Q_{market} = sign(P_0 - R)\sqrt{|P_0^2 - R^2|}$. (D.16)

$\Delta Q = Q_{market} - Q_{CAPM}$. (D.17)

$D_R = \Sigma_t tR_t/\Sigma_t R_t$. (D.18)

$D_Q = \Sigma_t tQ_t/\Sigma_t Q_t$. (D.19)

D.3 Test hypotheses

Hypothesis 1:

ΔQ predicts future return reversals or continuation after controlling for beta, size, value, and momentum. (D.20)

Hypothesis 2:

High D_Q stocks have stronger sensitivity to ERP shocks than low D_Q stocks. (D.21)

Hypothesis 3:

Positive ΔQ correlates with option-implied volatility, forecast dispersion, or narrative intensity. (D.22)

Hypothesis 4:

Negative ΔQ predicts credit stress, earnings disappointment, or valuation compression. (D.23)

Hypothesis 5:

Q/R explains valuation style differences better than P/E or price-to-book alone in selected regimes. (D.24)

D.4 Possible regression form

A simple test:

$FutureReturn_i = a + b_1\beta_i + b_2Value_i + b_3Momentum_i + b_4\Delta Q_i + \varepsilon_i$. (D.25)

If b_4 is significant after controls, ΔQ may contain information not captured by standard factors.

Another test:

$RateSensitivity_i = a + b_1 D_R,i + b_2 D_Q,i + \varepsilon_i$. (D.26)

If D_Q improves explanatory power, Q-duration may capture a real pressure-horizon effect.

Appendix E — Plain-English Glossary

Term	Meaning
A_t	Base-discounted cash-flow amplitude before CAPM risk rotation
R_t	Real admitted value; exactly the ordinary CAPM-discounted cash-flow value
Q_t	Imaginary CAPM-implied risk pressure; the orthogonal complement of R_t
θ_t	CAPM risk angle generated by βERP
q_t	Q_t/R_t, the risk-pressure ratio
Q_CAPM	Risk pressure implied by CAPM phase geometry
Q_market	Risk pressure implied by observed market price
ΔQ	Residual pressure gap between market-implied and CAPM-implied Q
D_R	Duration of admitted real value
D_Q	Duration of retained risk pressure
α_Q	Q-space residual, analogous to but distinct from CAPM alpha
α_θ	Angular residual between market-implied and CAPM-implied phase angle

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Appendix F — One-Paragraph Summary for Readers

CAPM normally converts beta risk into a higher required return, then DCF turns future cash flow into one real present value. The phase-twisted DCF model keeps the same CAPM value but reinterprets the process geometrically. First, each cash flow is discounted by a base rate to form an amplitude A_t. CAPM's beta premium then becomes a risk angle θ_t. The ordinary CAPM value R_t is the real projection $A_t \cos \theta_t$, while $Q_t = A_t \sin \theta_t$ is the orthogonal risk-pressure component that scalar DCF normally hides. This creates the identity $A_t^2 = R_t^2 + Q_t^2$ and allows new diagnostics such as Q_{market}, ΔQ, Q/R, and Q-duration. The result is not a replacement for CAPM, but a way to remember where CAPM risk went after discounting.

Appendix G — Admissible H-Mappings: From Mature Finance Valuations to Orthogonal Q-Coordinates

G.1 Why H-Mapping Needs Discipline

In the main article, Q was introduced through CAPM itself.

The CAPM-derived construction was:

$$A_t = CF_t/(1 + r_{base})^t. \quad (G.1)$$

$$R_t = CF_t/(1 + r_{CAPM})^t. \quad (G.2)$$

$$Q_t = \sqrt{(A_t^2 - R_t^2)}. \quad (G.3)$$

This was disciplined because Q_t was not freely invented. It was the orthogonal complement produced when CAPM risk-adjusted discounting was rewritten as a projection geometry.

But a natural question follows.

CAPM beta risk is not the only factor that can affect a security's value. Mature finance already has valuation methods for many other components:

real options,

liquidity discounts,

credit risk,

interest-rate sensitivity,

tax shields,

distress costs,

tail risk,

forecast dispersion,

factor premia,

and scenario-dependent payoffs.

So the question is:

If mature finance already gives a dollar valuation for a factor, can that valuation be used to construct a corresponding Q-coordinate? (G.4)

The answer is yes, but only with discipline.

A mature finance valuation gives a dollar effect. It does not automatically give an orthogonal Q-coordinate.

Therefore, the correct sequence is not:

factor $x_j \rightarrow$ arbitrary Q_j. (G.5)

The correct sequence is:

factor $x_j \rightarrow$ mature dollar valuation $\rightarrow$ residualization $\rightarrow$ geometric lift $\rightarrow Q_j$. (G.6)

In compact form:

$Q_j = \mathrm{Lift}_M[\mathrm{Residualize}(\mathrm{MatureValuation}_j(x_j) \mid R, Q_1,\ldots,Q_{j-1})]$. (G.7)

This is the central rule of Appendix G.

G.2 The Three-Layer Construction

An admissible H-map has three layers.

G.2.1 Layer 1 — Mature finance valuation

First, the factor must be valued by a mature finance method.

Let x_j be a factor.

Then:

$\Delta V_j = \mathrm{MatureValuation}_j(x_j)$. (G.8)

Examples:

ΔV_{option} = value of a real option. (G.9)

$\Delta V_{liquidity}$ = liquidity discount or liquidity premium effect. (G.10)

ΔV_{credit} = credit-risk adjustment. (G.11)

ΔV_{rate} = interest-rate shock value impact. (G.12)

ΔV_{tax} = tax shield value. (G.13)

$\Delta V_{distress}$ = expected distress cost. (G.14)

ΔV_{tail} = tail-loss measure such as VaR or Expected Shortfall. (G.15)

The first rule is:

No mature valuation, no admissible H-map. (G.16)

This prevents H from becoming an arbitrary conversion from narrative to money.

G.2.2 Layer 2 — Residualization

The mature valuation ΔV_j may overlap with R.

For example, if R already includes growth assumptions, then a separately computed growth option may double count part of R.

If R already uses a discount rate with a liquidity premium, then a separately computed liquidity Q may double count liquidity risk.

If R already includes credit-adjusted cash flows, then a separately computed credit Q may double count default risk.

Therefore the second layer is residualization:

$\Delta V_j^{\perp} = \text{Residualize}(\Delta V_j \mid R, Q_1,\ldots,Q_{j-1})$. (G.17)

The symbol $\perp$ means:

only the part not already admitted into R or earlier Q-components. (G.18)

The rule is:

No residualization, no orthogonality. (G.19)

This is the most important safeguard in Appendix G.

G.2.3 Layer 3 — Geometric lift

Once the residual dollar contribution is obtained, it must be converted into a Q-coordinate.

This final step is the geometric lift:

$Q_j = \text{Lift}_M(\Delta V_j^{\perp})$. (G.20)

The subscript M means the lift depends on the chosen metric or geometry.

In the simplest Euclidean case:

$P^2 = R^2 + Q^2$. (G.21)

For multiple Q-components:

$P^2 = R^2 + Q_1^2 + Q_2^2 + \ldots + Q_n^2$. (G.22)

But if Q-components overlap or are correlated, the safer form is:

$P^2 = R^2 + \vec{Q}^{T} G \vec{Q}$. (G.23)

where G is a metric matrix.

If:

$G = I$, (G.24)

then the Q-components are treated as orthogonal.

If:

$$G \neq I, \quad (G.25)$$

then the model admits covariance, overlap, or non-Euclidean weighting among pressure components.

G.3 Three Valid Types of H-Map

There is not one universal H-function.

Instead, there are several admissible H-map types.

The three most useful are:

direct component,

norm-preserving lift,

projection-complement lift.

G.3.1 Type A — Direct component H-map

The simplest construction is:

$$Q_j = \Delta V_j^{\perp}. \quad (G.26)$$

This says:

the residual dollar valuation is represented directly as a Q-coordinate. (G.27)

Example:

If a real option has residual value:

$$\Delta V_{option}^{\perp} = 5, \quad (G.28)$$

then:

$$Q_{option} = 5. \quad (G.29)$$

This is simple and intuitive.

But it does not preserve the Pythagorean price relation exactly.

If:

$$Z = R + iQ_{option}, \quad (G.30)$$

then:

$$|Z| = \sqrt{(R^2 + Q_{option}^2)}. \quad (G.31)$$

This is not equal to:

$$R + \Delta V_{option}^{\perp}. \quad (G.32)$$

Therefore Type A is useful for semantic accounting, but weaker for exact price reconstruction.

G.3.2 Type B — Norm-preserving lift

Suppose mature finance says that a factor adds or subtracts a residual dollar amount:

$$\Delta V_j^{\perp}. \quad (G.33)$$

Let M be the current valuation magnitude before adding this factor.

Then define:

$$Q_j = \operatorname{sign}(\Delta V_j^{\perp})\sqrt{|(M + \Delta V_j^{\perp})^2 - M^2|}. \quad (G.34)$$

This is the norm-preserving lift.

It constructs the Q-coordinate whose magnitude effect reproduces the mature valuation change.

If the factor is the first component beyond R, then:

$$M = R. \quad (G.35)$$

So:

$$Q_j = \operatorname{sign}(\Delta V_j^{\perp})\sqrt{|(R + \Delta V_j^{\perp})^2 - R^2|}. \quad (G.36)$$

This is useful when mature finance gives an additive component.

Examples include:

real option value,

tax shield value,

distress cost,

liquidity discount,

strategic premium,

or APV-style valuation components.

For small positive $\Delta V_j^{\perp}$:

$$Q_j \approx \sqrt{2M\Delta V_j^{\perp}}. \quad (G.37)$$

This approximation is important.

It shows that a small additive valuation premium may correspond to a larger orthogonal pressure coordinate.

So:

$\Delta V_j^{\perp}$ is the additive value effect. (G.38)

Q_j is the orthogonal coordinate that generates that value effect under the chosen geometry. (G.39)

They are not the same object.

G.3.3 Type C — Projection-complement H-map

Sometimes mature finance gives a before-and-after valuation.

Let:

A_j = value before factor adjustment. (G.40)

R_j = value after factor adjustment. (G.41)

Then:

Q_j = √(A_j² − R_j²). (G.42)

This is the same structure used in the CAPM construction.

It is best when a factor reduces admitted value from a larger base amplitude.

Examples:

risk-free value → CAPM-adjusted value; (G.43)

promised cash-flow value → credit-adjusted value; (G.44)

liquid value → illiquid value; (G.45)

base DCF value → distress-adjusted value; (G.46)

expected cash flow → certainty-equivalent cash flow. (G.47)

This is the cleanest construction for risk adjustments.

It says:

Q_j is the value-pressure removed from the real axis by the mature adjustment. (G.48)

G.4 Comparison of H-Map Types

H-map type	Input from mature finance	Q formula	Best use	Main warning
Direct component	residual value $\Delta V_j^{\perp}$	$Q_j = \Delta V_j^{\perp}$	semantic pressure accounting	not norm-preserving
Norm-preserving lift	additive value contribution	$Q_j = \text{sign}(\Delta V_j^{\perp})\sqrt{(M + \Delta V_j^{\perp})^2 - M^2}$		
Projection complement	before-and-after values	$Q_j = \sqrt{A_j^2 - R_j^2}$	discount-rate adjustments, credit risk, liquidity haircut, certainty equivalent	requires clear baseline A
Vector-metric version	multiple correlated components	$P^2 = R^2 + \vec{Q}^{\top} G \vec{Q}$	factor models and overlapping risks	G must handle covariance and overlap

This table is the core methodological guide.

It prevents the appendix from saying:

anything can become Q. (G.49)

Instead it says:

a factor can become Q only after mature valuation, residualization, and declared geometric lifting. (G.50)

G.5 Orthogonality Is Not Automatic

A crucial point must be stated plainly.

Mature finance valuations are not automatically orthogonal to R. (G.51)

A mature valuation gives a dollar effect.

But Q is an orthogonal or metric-adjusted coordinate.

Those are different concepts.

The rule is:

A factor can become Q_j only to the extent that it is not already admitted into R. (G.52)

Equivalently:

Q_j must represent residual value-pressure, not duplicated value already inside R. (G.53)

This means every Q_j depends on the definition of R.

If R is narrow, more factors can appear as Q.

If R is broad, fewer factors remain outside R.

For example:

If R is operating DCF excluding growth options, then real option value may become Q_option. (G.54)

If R already includes the growth option in projected cash flows, then Q_option should be zero or only the uncounted residual option value. (G.55)

Therefore:

R is the admitted-value boundary. (G.56)

Q is what remains outside that boundary but can still be valued. (G.57)

G.6 Orthogonality Check Table

Factor / valuation type	Mature valuation	Automatically orthogonal to R?	How to make it admissible

	exists?		
CAPM beta risk	Yes	No	define base amplitude A, CAPM R, then Q as projection complement
Real option value	Yes	No	exclude option from R, then lift residual option value
Liquidity discount	Yes	No	compare liquid baseline with illiquid adjusted value
Credit / default risk	Yes	No	compare promised or risk-free value with credit-adjusted value
Interest-rate duration	Yes	No	convert declared rate shock into residual value impact
VaR / Expected Shortfall	Yes	No	treat as downside pressure or negative Q-axis
Forecast dispersion	Indirectly	No	define R as consensus value and Q as valuation dispersion
Tax shield	Yes	No	treat APV tax shield as residual additive component
Expected distress cost	Yes	No	subtract from enterprise value, then lift as downside Q
Factor premia	Yes	No	residualize factors or use covariance metric G

The table shows the central lesson:

mature valuation is necessary but not sufficient. (G.58)

Orthogonality requires a declared protocol.

G.7 Methods for Residualization

There are several ways to compute:

$\Delta V_j^{\perp}$. (G.59)

The correct method depends on the purpose of the model.

G.7.1 Method 1 — Declared waterfall

The simplest method is a sequential waterfall.

Choose an order:

$R \to Q_1 \to Q_2 \to Q_3 \to \ldots \to Q_n$. (G.60)

Each new component is computed only after earlier components are already included.

For example:

R = operating DCF. (G.61)

Q_1 = CAPM risk pressure. (G.62)

Q_2 = residual real option pressure after CAPM. (G.63)

Q_3 = residual liquidity pressure after option value. (G.64)

This is easy to understand.

But it is order-dependent.

If the order changes, the components may change.

Therefore waterfall residualization is good for teaching, dashboards, and transparent modeling, but weaker for rigorous attribution.

G.7.2 Method 2 — Regression residualization

In data-rich settings, a factor can be residualized statistically.

Write:

$x_j = \text{Explained}(x_j \mid \text{existing factors}) + x_j^{\perp}$. (G.65)

Then use only the residual part:

$Q_j = H_j(x_j^{\perp})$. (G.66)

This is useful when dealing with factors such as:

beta,

size,

value,

momentum,

quality,

liquidity,

volatility,

duration,

credit spread,

or macro exposures.

The logic is:

only the unexplained part of the factor becomes a new Q-coordinate. (G.67)

This reduces double counting.

G.7.3 Method 3 — Shapley-style allocation

When factors interact strongly, a fixed order may be unfair.

Let V(S) be the valuation produced by including a set S of factors.

Then factor j's marginal contribution depends on which other factors are already included:

$V(S \cup \{j\}) - V(S)$. (G.68)

A Shapley-style allocation averages this marginal contribution across possible sets S.

Conceptually:

$\Delta V_j^{Shapley} = average_S [V(S \cup \{j\}) - V(S)]$. (G.69)

This gives a fairer allocation when factor effects overlap.

It is useful for:

real options plus growth assumptions,

tax shield plus distress costs,

liquidity plus credit risk,

platform optionality plus forecast dispersion,

or multi-factor valuation.

After Shapley allocation, the result can be lifted into Q:

$Q_j = Lift_M(\Delta V_j^{Shapley})$. (G.70)

This method is more complex, but more defensible when interactions matter.

G.7.4 Method 4 — Metric geometry

Sometimes the components should not be forced to be orthogonal.

In that case, use:

$P^2 = R^2 + \vec{Q}^{T} G \vec{Q}$. (G.71)

Here:

$\vec{Q}$ = vector of pressure coordinates. (G.72)

G = metric matrix. (G.73)

If G has off-diagonal terms, then Q-components interact.

This is appropriate when factor pressures are correlated.

For example:

liquidity risk and credit risk may rise together; (G.74)

volatility and funding stress may rise together; (G.75)

growth optionality and duration risk may overlap; (G.76)

momentum and narrative pressure may overlap. (G.77)

The metric form is more mature than pretending all Q-components are perfectly orthogonal.

G.8 Examples of Mature-Valuation-Based H-Maps

G.8.1 Real option value

Suppose mature real-options analysis gives:

ΔV_{option} = option value of expansion, abandonment, deferral, or strategic flexibility. (G.78)

If R excludes this option, then the residual is:

$\Delta V_{option}^{\perp} = \Delta V_{option}$. (G.79)

Using norm-preserving lift:

$Q_{option} = \sqrt{[(R + \Delta V_{option})^2 - R^2]}$. (G.80)

If R already includes the option value through optimistic cash-flow projections, then:

$\Delta V_{option}^{\perp} = 0$, or only the uncounted residual option value. (G.81)

The rule is:

real option value can become Q_{option} only if it is not already admitted into R. (G.82)

G.8.2 Liquidity discount

Let:

V_{liquid} = valuation assuming normal liquidity. (G.83)

$V_{illiquid}$ = valuation after liquidity adjustment. (G.84)

Then a projection-complement construction gives:

$Q_{liquidity} = \sqrt{(V_{liquid}^2 - V_{illiquid}^2)}$. (G.85)

This treats liquidity as a pressure that prevents the liquid valuation from being fully admitted as real value.

If R already equals $V_{illiquid}$, then $Q_{liquidity}$ is the complement relative to V_{liquid}.

If R already includes a liquidity premium in the discount rate, a second liquidity Q should not be added unless it represents residual liquidity pressure beyond that adjustment.

G.8.3 Credit / default risk

Let:

$V_{promised}$ = risk-free value of promised cash flows. (G.86)

V_{credit} = credit-adjusted value. (G.87)

Then:

$Q_{credit} = \sqrt{(V_{promised}^2 - V_{credit}^2)}$. (G.88)

This is analogous to CAPM Q.

The promised value is the amplitude-like baseline.

The credit-adjusted value is the admitted real value.

The credit Q is the pressure removed by default risk.

G.8.4 Interest-rate sensitivity

Duration and DV01 already convert rate changes into dollar value impacts.

A simple rate-shock approximation is:

$\Delta V_{rate} \approx -\text{Duration} \times V \times \Delta y$. (G.89)

Or:

$\Delta V_{rate} \approx \text{DV01} \times \text{basis-point shock}$. (G.90)

This gives a mature dollar value impact.

To convert into Q, use a signed norm-preserving lift:

$Q_{rate} = \text{sign}(\Delta V_{rate})\sqrt{|(R + \Delta V_{rate})^2 - R^2|}$. (G.91)

This is not a claim that rate exposure is automatically orthogonal.

It becomes a Q-coordinate only under a declared shock scenario and residualization rule.

G.8.5 Tail risk

VaR and Expected Shortfall already produce dollar loss measures.

Let:

ES_α = expected shortfall at confidence level α. (G.92)

A downside-pressure construction may use:

$Q_{tail} = -\sqrt{|R^2 - (R - ES_\alpha)^2|}$. (G.93)

The negative sign indicates downside pressure.

This should be handled carefully.

Tail risk is not ordinary upside optionality.

It is directional loss pressure.

Therefore it may belong on a separate negative Q-axis, or in a signed Q convention.

G.8.6 Forecast dispersion

Suppose analyst forecasts produce a distribution of DCF values:

$V_1, V_2, \ldots, V_n$. (G.94)

Let:

R = mean or consensus valuation. (G.95)

Then valuation dispersion may be:

σ_V = standard deviation of $\{V_1, V_2, \ldots, V_n\}$. (G.96)

A simple direct component map is:

$Q_{dispersion} = \sigma_V$. (G.97)

A tail version could use:

Q_dispersion-tail = ExpectedShortfall of valuation downside distribution. (G.98)

This is admissible if R is explicitly defined as the consensus or expected value.

Then Q_dispersion represents uncertainty around that admitted mean.

G.8.7 Tax shield and distress cost

In adjusted present value, or APV-style thinking, firm value can be decomposed into:

$V_{firm} = V_{unlevered} + PV_{tax_shield} - PV_{distress_cost}$ + other adjustments. (G.99)

The additive components can be converted by norm-preserving lift.

For tax shield:

$Q_{tax} = \sqrt{[(R + PV_{tax_shield}^{\perp})^2 - R^2]}$. (G.100)

For distress cost:

$Q_{distress} = -\sqrt{|R^2 - (R - PV_{distress_cost}^{\perp})^2|}$. (G.101)

The signs matter.

Tax shield may be positive pressure.

Distress cost is downside pressure.

Both are admissible only after checking whether R already includes them.

G.8.8 Factor premia

For a factor model:

$r = r_{base} + \beta_1\lambda_1 + \beta_2\lambda_2 + \dots + \beta_n\lambda_n$. (G.102)

Each factor premium $\beta_k\lambda_k$ may be converted into a factor pressure.

A single-angle version uses:

$\Lambda = \beta_1\lambda_1 + \beta_2\lambda_2 + \dots + \beta_n\lambda_n$. (G.103)

$\cos\theta_t = [(1 + r_{base})/(1 + r_{base} + \Lambda)]^t$. (G.104)

$Q_t = A_t \sin\theta_t$. (G.105)

A multi-axis version uses:

$P^2 = R^2 + \vec{Q}^{\dagger} G \vec{Q}$. (G.106)

where $\vec{Q}$ contains factor-specific pressure coordinates.

Because factors are often correlated, G should not automatically be assumed to be the identity matrix.

G.9 General Admissibility Checklist

Before adding any H-map, ask the following questions.

Requirement	Question
Mature valuation basis	Which accepted finance method produces ΔV_j?
Dollar anchor	What cash flow, asset value, exposure, or portfolio value anchors the dollar amount?
Boundary of R	What value is already admitted into R?
Residualization rule	What overlap with R and prior Q-components is removed?
Lift type	Direct, norm-preserving, projection-complement, or metric-vector?
Sign convention	Is Q positive, negative, symmetric, or directional?
Time horizon	Over what horizon is the factor valued?
Metric	Are Q-components orthogonal, correlated, or covariance-adjusted?
Calibration	Can the H-map be estimated from prices, spreads, forecasts, options, losses, or historical data?
Falsifiability	Does Q_j improve explanation, prediction, diagnosis, or classification?

A factor should not become Q_j unless these questions have explicit answers.

G.10 The General H-Map Formula

The mature form of H can now be written as:

$Q_j = H_j(x_j)$. (G.107)

But this should be understood as shorthand.

The full form is:

$Q_j = \text{Lift}_M[\text{Residualize}(\text{MatureValuation}_j(x_j) \mid R, Q_1,\ldots,Q_{j-1})]$. (G.108)

Equivalently:

$Q_j = \text{Lift}_M(\Delta V_j^{\perp})$. (G.109)

where:

$\Delta V_j = \text{MatureValuation}_j(x_j)$. (G.110)

$\Delta V_j^{\perp}$ = residual value contribution not already admitted into R. (G.111)

This makes H a protocol, not a magic function.

In plain English:

H does not create value from a factor. H converts a mature residual valuation into a declared pressure coordinate. (G.112)

G.11 The Most Important Boundary Rule

The strongest rule is:

R defines what is already admitted. Q defines what is valued but not admitted into R. (G.113)

Therefore, every model must first declare R.

For example:

R = CAPM DCF value. (G.114)

R = operating DCF excluding real options. (G.115)

R = consensus analyst DCF. (G.116)

R = credit-adjusted bond value. (G.117)

R = liquid-market comparable value. (G.118)

Different choices of R create different Q spaces.

This is not a weakness if declared clearly.

It is a governance rule.

Without a declared R-boundary, Q becomes ambiguous.

G.12 Closing Summary

Mature finance already contains many ways to value non-price factors in dollars.

These include:

real option valuation,

liquidity discounts,

credit adjustments,

duration and DV01,

tail-risk measures,

forecast dispersion,

tax shields,

distress costs,

and factor premia.

The Q-framework does not replace those methods.

It uses them as valuation engines.

But their outputs are not automatically orthogonal to R.

To become Q-coordinates, they must be:

valued by a mature method,

residualized against admitted value,

lifted into a declared geometry,

signed correctly,

horizon-defined,

and tested where possible.

The final rule is:

Mature finance supplies ΔV. The Q-framework supplies the geometry. Residualization prevents double counting. The lift converts valuation contribution into pressure coordinate. (G.119)

Or in one line:

An admissible H-map is not a free conversion from factor to dollars; it is a disciplined protocol that takes a mature finance valuation, removes overlap with admitted value R, and lifts the remaining residual into a declared Q-coordinate. (G.120)

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